

INDC International Nuclear Data Committee

TABLE OF NUCLEAR ELECTRIC QUADRUPOLE MOMENTS

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ABSTRACT

This Table is a compilation of experimental measurements of static electric quadrupole moments of ground states and excited states of atomic nuclei throughout the periodic table. To aid identification of the states, their excitation energy, half-life, spin and parity are given, along with a brief indication of the method and any reference standard used in the particular measurement. Experimental data from all quadrupole moment measurements actually provide a value of the product of the moment and the electric field gradient (EFG) acting at the nucleus. Knowledge of the EFG is thus necessary to extract the quadrupole moment. A single recommended value of the moment is given for each state, based, for each element, wherever possible, upon a standard reference moment for a nuclear state of that element studied in a situation in which the electric field gradient has been well calculated. For several elements one or more subsidiary reference EFG/moment references are required and their use is specified.

The literature search covers the period to April 2021.

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INTRODUCTION AND MOTIVATION

The electric quadrupole moment is a feature of atomic nuclei that finds relevance in a wide range of fields of research and more general measurement. The quadrupole moment is most obviously of value in the study of nuclei themselves, being a measure of the deviation of the nuclear charge distribution from a spherical shape. However, in addition, measurement of nuclear quadrupole interactions forms a vital investigatory tool in many areas of condensed matter physics extending into the chemical, biological and medical sciences, aiding the elucidation of structures of complex molecules and their function.

Establishing accepted values of the quadrupole moment of any nuclear state faces a fundamental problem. The difficulty lies in the fact that the measurable quantity is the interaction energy of the quadrupole moment, eQ , with its surroundings, usually written as $h\nu_Q = e^2qQ$. This interaction involves the product of the moment with the electric field gradient (EFG), eq , acting at the nucleus. The situation can be contrasted to the challenge of measuring the nuclear magnetic dipole moment, μ , which similarly involves its product with the magnetic field B acting at the nucleus, $h\nu_M = \mu B$. In the magnetic case it is a relatively simple matter to apply a known magnetic field, generated by a suitable arrangement of coils carrying a precisely measured current, which can be varied at the will of the experimenter. The nuclear magnetic moment can be extracted from this product, and hence measured, by a wide range of methods.

However, it is not possible to generate EFGs in the laboratory which are sufficiently large to produce measurable energy differences. Given the small scale of nuclear quadrupole moments, of order 1 barn [100 fm^2], to provide an energy splitting between levels of order 10 MHz requires an EFG of order 10^{17} Vcm^{-2} . Such large field gradients are indeed to be found at nuclei when in atoms, ions or molecules and in non-cubic solids. The field gradient cannot be determined separately but must be obtained by calculation. Difficulties in calculating the EFG from first principles have historically set serious limitations on our knowledge of nuclear electric quadrupole moments.

This is not the place to describe the development of multi-electron electric field distribution theory, however, until the advent of more powerful computers only the simplest atoms or ions, with few electrons, were open to accurate calculation. For this reason, the study of quadrupole hyperfine interactions in muonic atoms, in which the muon wavefunction (and hence charge distribution and EFG) at the nucleus can be accurately predicted, was early seen as a fruitful way to establish reliable values of quadrupole moments, especially in heavier elements where the interaction can be readily resolved. For multi-electron atoms and ions, many delicate features of the calculation were beyond the power of computation until the past two decades or so. Prior to this, the best EFG estimates involved approximations and corrections related to the distortion of inner closed electron shells, collectively associated with the name Sternheimer shielding (or anti-shielding) factors, which could be uncertain to 10% or more. Thus, although experimentally the nuclear quadrupole interaction could be measured to considerably better precision, the nuclear quadrupole moments extracted from the measurements were hard to pin down.

In recent years multi-electron computation has advanced to the extent that it is now possible to make calculations of EFGs in most atoms and ions, and also in many molecules and solids. The results show admirable self-consistency and are proving reliable to a degree comparable to that of the best muonic atom results. First in 1992 and again in 2001, 2008 and 2018 Pekka Pyykko, one of the pioneers in such calculations, has published an authoritative listing of recommended values of the quadrupole moments of selected, usually stable, nuclear states in the majority of elements,

including those most commonly used in applications beyond the realm of nuclear structure physics. Adopting such a set of standard reference values allows measurements using the reference state to be used to calibrate EFGs in other environments which, by extension, allows moments of other isotopes, and excited states in all isotopes, to be related to the single reference moment. The modern calculations include all effects previously grouped under the name of shielding.

There exist in the literature many excellent experimental results giving rise to values of nuclear quadrupole moments. Each measurement required the authors to specify the EFG at the nucleus in their particular experimental system. The published Q values depend upon the individual choice of EFG and can thus be difficult to compare consistently without considerable effort. This is especially true if the experiments were published many years apart, during which time the reference value(s) adopted may have changed. Only recently has it been possible, with the aid of much improved EFG calculations, to prepare a tabulation in which the adopted standards are likely to prove stable over at least the medium term, trustworthy in many cases to a few percent or better.

The Table herein presents a listing of measured quadrupole moments normalized, as far as possible, to a set of identified reference moments, one or more for each element, which have been derived from measurement in a situation in which the EFG has been calculated either by a modern computation or in a simpler system such as a muonic atom. There are still some 17 elements, He, Si, P, Ar, Ag, Cd, Te, Ce, Tm, W, Pt, Tl, Po, At, Cm, Bk and Cf for which no such clear reference standard exists. For the majority of the remaining elements, the reference values chosen by Pyykko in his most recent listing have been adopted [1]. The few exceptions are noted in the Table.

It is frequently not possible to normalize all measurements to the adopted reference moment for an element. This arises most often in connection with excited state measurements when an equivalent measurement on the reference (usually stable) isotope in the same conditions is not possible. In such cases a secondary standard EFG has often been adopted. These are briefly specified in the Table in the heading for each element.

In contrast to the forgoing discussion there do exist less widely available and less accurate methods of measurement of the quadrupole moment which are free of uncertainty of estimation of an associated EFG. These involve calculation of field gradients provided by free electric charges and thus nothing more than Coulomb's law. They include electron scattering (ES) and re-orientation of the angular distribution during coulomb excitation (CER) both of which have been extensively used in the study of short lived excited nuclear states. Calculation of the electromagnetic interaction of the charged projectile with the nucleus is straightforward in principle, and the quadrupole moment can be readily extracted, although the experiments have limited accuracy. Moments determined using these methods are therefore not directly related to, or dependent upon, the value of the adopted standard quadrupole moment for that element.

This Table updates that issued in 2013 [2]; the most recent entries were published in early 2021.

POLICIES

Signs

Signs are indicated when the sign can be determined from experimental data. Where the sign is not given by the measurement, no sign is indicated in the Table, although it can sometimes be inferred either from systematics or from the magnitude of the result.

Results and Uncertainties

Experimental values and their associated errors are as given by the authors subject to a policy of limiting significant figures. Numerical errors with digits above 15 have, in most cases, been rounded to 2 and errors with first digit 2 or greater rounded down, if the second digit is 4 or lower, otherwise rounded up. Results have also been rounded to give no more significant figures than the rounded error would allow. Thus, a published value 0.953(65) has been rounded to 0.95(7) and 0.25(16) rounded to 0.3(2).

Electric Quadrupole Moments

These are listed in units of barns ($1 \text{ b} = 10^{-28} \text{ m}^2$). As explained in the introduction, with modern computations of the EFG, corrections relating shielding caused by polarization of atomic electrons, known as Sternheimer Corrections, are no longer needed. Where there is more than one reference isotope this is possible without causing any confusion since moment ratios are frequently determined to far greater precision than their individual values. References to original publication are given, both in the form of the Nuclear Structure Reference listing maintained by the National Nuclear Data Centre, Brookhaven National Laboratory, and the journal reference. A listing of journal abbreviations is given in Appendix 3.

Reference moments and EFGs.

Justification of the reference values should be sought in the reference(s) given, usually Pyykko [1], or, for the secondary standards, the measurements in which they have been used.

Recommended Values.

The Table gives, for each listed state, identified by energy [in keV], half-life and spin, a single value for the quadrupole moment. The state data are taken from the most recent ENSDF files available in the ENSDF database, National Nuclear Data Center, BNL. The method involved in the quadrupole measurement is identified by one of the abbreviations listed in Appendix 2. Entries have been normalized to the given value of the reference isotope moment wherever possible, so values will in general differ from those in the original publications. Where several experimental results exist for the same state, attempts to make valid weighted averages have been avoided since frequently there exist non-statistical differences between experiments that cannot be taken into account. Most often one of the more recent results, with a relatively small estimated uncertainty, is given. Readers who require a full listing of all results on a given state should consult the most recent version of the extended Table [3].

New Entries, Significant Changes and Problem Elements.

As well as adjustments to several hundred existing results in the light of new field gradient calculations, the Table includes many entirely new measurements since the 2013 version. These extend the range of known quadrupole moments for isotopes of Na, Mg, Al, Ca, Mn, Cu, Rb, Cd,

Bi, Po, At, Fr, Ac and No. The in-source and collinear laser spectroscopy methods are the major sources. 2020 saw the welcome return of muon spectroscopy to the area of quadrupole moment determination with the report of results on Re isotopes. Detailed analysis of the limited-resolution, multi-component muon X-ray spectrum led to new moments for $^{185,187}\text{Re}$, reduced by 9.5% compared to the previous standards.

The far fewer new excited state measurements have been mainly by the Coulomb excitation re-orientation method. Noted additions are on levels in Se, Sr, Sn, Te, I, and Po isotopes.

Many elements have seen changes in their reference moments since the last Table appeared in 2013. These reflect improvements in both experiment and EFG calculation. In many cases the adjustments are at the few % level, however, there have been some major changes. These are Zn (-19%), Sb (+51%), Fr (+10%) and Th (-28%).

In several elements, errors have been reduced, most markedly for the deuteron for which Q is now determined to 0.01%. Otherwise, minimum uncertainties are typically a few % but several are far greater (V 19%, Cr 33%, Co 7%, Ni, 9%, Cu 7%, Zn 8%, Nb 6%, Nd 10%, Pm 27%, Sm 10%, Pb 61%, Rn 10%, Fr 10%, Pa 12%, Am 31% and Es 12%).

Not all new results provide clarification for a Table such as this. There are difficulties in reconciling apparently equally good results for Ca and Bi moments from different methods of either measurement or calculation.

The subject is not closed as work continues.

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- [2] N.J. Stone, IAEA Vienna Report INDC(NDS)-0650 (2013). [IAEA Nuclear Data Services](#)
- [3] N.J. Stone, IAEA Vienna Report INDC(NDS)-0732 (2017). [IAEA Nuclear Data Services](#)

APPENDIX 1 - Explanation of the Table

The Table gives information in columns as follows:

Element	Identifies the element for the following isotopes and specifies the reference isotope(s) for that element. Also, in the first row for each element, the system in which the EFG has been calculated to obtain the reference moment(s) is given briefly. Secondary standards are listed in subsequent rows and identified by Letters A,B,C For more detail on the EFG calculations see [1] above.
Nucleus	Identifies the nucleus by mass number A and atomic number Z , with its chemical symbol. Nuclei are grouped by element in increasing sequence of atomic number and by increasing mass number for each element.
$E(\text{level})$	Gives the energy of the state on which the measurement is made, rounded to the nearest kilovolt, 0 being the ground state. Where placement of the level with respect to the ground state is unknown, this is denoted by addition of an offset x or y .
$T_{1/2}$	Gives the half-life $T_{1/2}$ of the state: Units y = years, d = days, h = hours, m = minutes, s = seconds, ms = milliseconds (10^{-3} s), μs = microseconds (10^{-6} s), ns = nanoseconds (10^{-9} s), ps = picoseconds (10^{-12} s) and fs = femtoseconds (10^{-15} s).
I^π	Gives the spin (I) and parity (π) of the state. Uncertain values are given in brackets.
$Q(b)$	Gives the measured nuclear electric quadrupole moment Q in units of the barn ($1 \text{ barn} = 10^{-28} \text{ m}^2$). No sign is given if it was not determined by the experiment. The uncertainty in the result is given in brackets, subject to the policy declared in the introduction. Thus $+1.27(10)$ means a value of $+1.27$ barns with uncertainty 0.10 barns.
<i>Ref. Std.</i>	In this column the reference standard upon which the listed result depends is given. There is no entry when the method used does not depend upon an adopted standard (i.e. a Coulomb Excitation Reorientation (CER) measurement).
<i>Method</i>	The method used in the measurement is briefly identified here. A list of abbreviations used is given below. In view of the great proliferation of specialized methods, this method description is limited and for detailed information reference should be made to the original publication. Re-evaluation of the published result, where it involves re-normalization to the adopted value of the reference standard by the tabulator, is not indicated specifically.
<i>References</i>	1. The NSR keyword reference is given. These can be further identified by reference to the Brookhaven National Nuclear Data Centre website www.nndc.bnl.gov. 2. The original journal or other publication reference is given wherever possible. 3. Unpublished results are included only where they offer a major contribution and no refereed publication exists.

APPENDIX 2 - Experimental Method Abbreviations

A	Atomic spectroscopy
AB	Atomic beam magnetic resonance
AB/MS	Atomic beam and molecular spectroscopy
ABLDF	Atomic beam with laser double resonance detection
ABLFS	Atomic beam with laser fluorescence spectroscopy
ABLS	Atomic beam laser spectroscopy
β -NMR	NMR of in-beam polarized nuclei with beta asymmetry detection
β -NQR	Nuclear quadrupole resonance with beta detection
β -RadOP	Beta-ray detection of optical pumping
B(E2)	Value based on measured E2 transition probability
CER	Coulomb excitation reorientation
CERP	Precession of coulomb excitation reorientation
CFBLS	Collinear fast beam laser spectroscopy
CFBLS/ β -NMR	Collinear fast beam laser spectroscopy: NMR with beta detection
CLS	Collinear laser spectroscopy
EPR	Electron paramagnetic resonance
ES	Electron scattering
IGLIS	In gas laser ionization spectroscopy
IPAC	Integral perturbed angular correlation
ISLS	In source laser spectroscopy
LEMS	Level mixing spectroscopy
LRFS	Laser resonance fluorescence spectroscopy
LRIS	Laser resonance ionization spectroscopy
LS	Laser spectroscopy
MA	Microwave absorption in gases
MAPON	Modulated adiabatic passage NMR on oriented nuclei
MB	Molecular beam magnetic resonance
ME	Mossbauer effect
MS	Molecular spectroscopy
Mu-X	Muonic X-ray hyperfine structure
NMR/ON	Nuclear magnetic resonance on oriented nuclei
NO/ME	Mossbauer effect on oriented nuclei
NO/S	Static nuclear orientation with gamma detection
NQR	Nuclear quadrupole resonance
NSLR	Nuclear spin-lattice relaxation
O	Optical spectroscopy
OD	Optical double resonance
OP/RD	Optical pumping with radiative detection
PAC	Perturbed angular correlation
Pi-X	Pionic X-ray Hyperfine Structure
Q	Quadrupole resonance
QI-NMR/ON	Quadrupole interaction resolved NMR on oriented nuclei
QIR	Quadrupole Interaction deduced from Relaxation Time
R	Re-evaluated data, or (for revised reference standard) adjusted by tabulator

RIMS/LS	Resonant ionization mass spectrometry/laser spectroscopy
TDPAC	Time-dependent perturbed angular correlation
TDPAD	Time-dependent perturbed angular distribution
TF	Transient field
TFLD	Tilted foil time-differential perturbed gamma angular distribution
TLS	Trap laser spectroscopy

APPENDIX 3 - Literature Reference Abbreviations

Acta Phys Pol	Acta Physica Polonica
ADNDT	Atomic Data and Nuclear Data Tables
AECL	Report of Atomic Energy of Canada Ltd.
ARHMI	Annual Report of the Hahn Meitner Institute, Berlin
BAPS	Bulletin of the American Physical Society
Bk88	Nuclei Far From Stability, AIP Conf. 164 Rosseau Lake, Canada 1988
Can J Chem	Canadian Journal of Chemistry
CERN EP	CERN EP Division Report
Chem Phys Lett	Chemical Physics Letters
Chin Phys Lett	Chinese Physics Letters
CPL	Chemical Physics Letters
CzJP	Czech Journal of Physics
Eur Phys J	European Physics Journal
Europhys Lett	Europhysics Letters
HP Ac	Helvetia Physica Acta
HFI	Hyperfine Interactions
IAN Ser Fiz	Izv. Akad. Nauk. SSSR Ser Fiz (trans Bull. Acad. Sci. USSR, Phys. Ser.)
IoP Conf	Institute of Physics Conference series
J Phys	Journal of Physics (London)
J Phys Radium	Journal de Physique et Radium (Paris)
JCP	Journal of Chemical Physics
JINC	Journal of Inorganic and Physical Chemistry
JPCR	Journal of Physical and Chemical Reference Data
JPSJ	Journal of the Physical Society of Japan
JPPa	Journal de Physique (Paris)
LINPR	Leningrad Inst. of Nuclear Physics Report
Mol Phys	Molecular Physics
Nat Commun	Nature Communications
NIM	Nuclear Instruments and Methods
NIMPR	Nuclear Instruments and Methods in Physics Research
NP	Nuclear Physics
ORNL	Oak Ridge National Laboratory Report
Ospk	Opt. Spektrosk. (trans. Optics and Spectroscopy (USSR))
PC	Private Communication
Phca	Physica
PhMg	Philosophical Magazine
Phys Scr	Physica Scripta
PL	Physics Letters
PR	Physical Review
PRep	Physics Reports
PRL	Physical Review Letters
PRS	Proceedings of the Royal Society of London
RIKEN	Report of the RIKEN laboratory, Japan
Sol St Comm	Solid State Communications
STMP	Springer Texts in Modern Physics

TH	Thesis B.R. Casserberg, Princeton, 1968
Theor Chem Acta	Theoretical Chemistry Acta
UCRL	University of California Lawrence Berkeley Report
UkrF	Ukraine Fiz. Zh.
YadF	Yadern Fys (trans. Soviet Journal of Nuclear Physics)
ZNat	Zeitschrift fur Naturforschung
ZP	Zeitschrift fur Physik

APPENDIX 4

TABLE OF NUCLEAR ELECTRIC QUADRUPOLE MOMENTS

Element	Isotope	E(level)	T _{1/2}	I*	Q(b)	Ref. Std.	Method	NSR Keynumber	Journal reference
Hydrogen		<i>Efg at the deuterium nucleus calculated for HD and D2</i>						2010Pa42	PR A81 042526 (2010)
Reference isotope	1 H 2	0	stable	1+	+0.0028578(3)		MB,R	1958Qu02 2010Pa42	PR 112 1929 (19) PR A81 042526 (2010)
Lithium		<i>Efg at Li in the LiH molecule</i>							
	3 Li 6	0	stable	1+	-0.000806(6)	[Li7]	MB	2005Bo45 1998Ce04	PR C72 044309 (2005) PR A57 2539 (1998)
Reference isotope	3 Li 7	0	stable	3/2-	-0.0400(3)		MB	2008Py02 1998Ce04	Mol Phys 106 1965 (2008) PR A57 2539 (1998)
	3 Li 8	0	840 ms	2+	+0.0314(2)	[Li7]	β-NMR	2005Bo45	PR C72 044309 (2005)
	3 Li 9	0	178 ms	3/2-	-0.0304(2)	[Li7]	β-NMR	2011Av08	J Phys G 38 075102 (2011)
	3 Li 11	0	8.75 ms	3/2-	(-)0.0333(5)	[Li7]	β-NMR	2008Ne11	PRL 101 132502 (2008)
Beryllium		<i>Calculation of the quadrupole coupling constant for the 3P2 state of the Be atom</i>							
Reference isotope	4 Be 9	0	stable	3/2-	+0.0529(4)		AB	1991Su05	CPL 177 91 (1991)
	4 Be 10	3368	0.125 ps	2+	-0.08(7)		CER	2012Or05	PR C86 041303 (2012)
Boron		<i>Calculation of the quadrupole coupling constant of the 2P3/2 ground state of the B atom</i>							
	5 B 8	0	0.77s	2+	+0.0643(14)	[11B]	β-NQR	2006Su13	PR C74 024327 (2006)
	5 B 10	0	stable	3+	+0.0846(2)	[11B]	AB	2008Py02 1970Ne21	Mol Phys 106 1965 (2008) PR A2 1208 (1970)
Reference isotope	5 B 11	0	stable	3/2-	+0.04059(10)		AB	2008Py02 1970Ne21	Mol Phys 106 1965 (2008) PR A2 1208 (1970)
	5 B 12	0	20.2 ms	1+	0.0132(3)	[11B]	β-NMR	1992Mi18	PRL 69 2058 (1992)
	5 B 13	0	17.3 ms	3/2-	(+)0.0365(8)	[11B]	β-NMR	2004Na38	NP A746 509c (2004)
	5 B 14	0	12.5 ms	2-	0.0297(8)	[11B]	β-NMR	1996Iz01	PL B366 51 (1996)
	5 B 15	0	9.9 ms	3/2-	0.0379(11)	[11B]	β-NMR	1996Iz01	PL B366 51 (1996)
	5 B 17	0	5.1 ms	(3/2-)	0.0385(15)	[11B]	β-NMR	2003Og03	PR C67 064308 (2003)
Carbon		<i>Calculation of the quadrupole coupling constant of the 3P2 state of the C atom</i>							
Reference isotope	6 C 11	0	20.4 m	3/2-	0.0333(2)		AB	2008Py02 1969Sc34	Mol Phys 106 1965 (2008) PR 181 137 (1969)

	6 C 12	4438	45 fs	2+	+0.06(3)	[11C]	CER	1983Ve01	PL B122 23 (1983)
Nitrogen	<i>Calculation of the quadrupole coupling constant of the 1P1 state of the N+ ion</i>								
	7 N 12	0	11.0 ms	1+	+0.0100(9)	[14N]	β -NMR	1998Mi10	PL B420 31 (1998)
<i>Reference isotope</i>	7 N 14	0	stable	1+	+0.02044(3)		AB/MS	2008Py02 1997To06	Mol Phys 106 1965 (2008) CPL 265 60 (1997)
	7 N 16	0	7.13 s	2-	(-)0.018(2)	[14N]	β -NMR	2001Ma42	PRL 86 3735 (2001)
	7 N 18	0	624 ms	1-	(+)0.0123(12)	[14N]	β -NMR	1999Og03	PL B451 11 (1999)
Oxygen	<i>Calculation of the quadrupole coupling constant of the 3P2 state of the O atom</i>								
	8 O 13	0	8..6 ms	3/2-	0.0111(8)	[17O]	β -NQR	1999Ma46	PL B459 81 (1999)
<i>Reference isotope</i>	8 O 17	0	stable	5/2+	-0.0256(2)		R/EPR	2018Py01 1969Sc34	Mol Phys 116 1328 (2018) PR 181 137 (1969)
	8 O 18	1982	1.94 ps	2+	-0.036(9)	[17O]	CER	1983Gr28	NP A411 329 (1983)
	8 O 19	0	26.9 s	5/2+	0.00362(13)	[17O]	β -NMR	1999Mi16	PL B457 9 (1999)
Fluorine	<i>Calculation of the quadrupole coupling constant of the F2 molecule</i>								
	9 F 17	0	64.5 s	5/2+	0.076(4)	[19F 197keV]	β -NMR	1974Mi21	NP A236 415 (1974)
	9 F 18	1121	162 ns	5+	0.071(6)	[19F 197keV]	β -NMR	1974Mi21	NP A236 415 (1974)
<i>Reference isomer</i>	9 F 19	197	89.3 ns	5/2+	-0.0942(9)		PAC	2008Py02	Mol Phys 106 1965 (2008)
	9 F 20	0	11.1 s	2+	0.056(4)	[19F 197keV]	β -NMR	1974Mi21	NP A236 415 (1974)
	9 F 21	0	4.16 s	5/2+	0.011(2)	[19F 197keV]	β -NMR	1999Mb13	HFI 120/121 673 (1999)
	9 F 22	0	4.23 s	4+	0.003(2)	[19F 197keV]	β -NMR	2010Mi13	NP A834 75c (2010)
Neon	<i>Calculation of the quadrupole coupling constant of the 3P2 state of the Ne atom</i>								
	10 Ne 20	1634	0.73 ps	2+	-0.23(3)		CER	1981Sp07	PRep 73 369 (1981)
<i>Reference isotope</i>	10 Ne 21	0	stable	3/2+	+0.1016(8)		O/AB	2018Py01 1972Du06	Mol Phys 116 1328 (2018) PR A5 1036(1972)
	10 Ne 22	1275	3.6 ps	2+	-0.215(12)		CER	2018He12	PL B782 468 (2018)
	10 Ne 23	0	37.2 s	5/2+	0.145(13)	[21Ne]	CFBLS	2005Ge06	PR C71 064319 (2005)

Sodium		Calculation of the quadrupole coupling constant of the 3P state of the Na atom							
Reference isotope	11 Na 20	0	0.448 s	2+	+0.101(8)	[23Na]	β -NMR	2009Mi04	PL B672 120 (2009)
	11 Na 21	0	22.5 s	3/2+	0.138(11)	[23Na]	β -NMR	2009Mi04	PL B672 120 (2009)
	11 Na 22	0	2.60 y	3+	+0.180(11)	[23Na]	β -NMR	1998Ga44	Eur Phys J A3 313 (1998)
	11 Na 23	0	stable	3/2+	+0.104(1)		O	2008Py02 2006Da14	Mol Phys 106 1965 (2008) J Phys B 39 3111 (2006)
	11 Na 25	0	59.1 s	5/2+	0.0015(3)	[23Na]	β -NMR	2004Og13	HFI 159 235 (2004)
	11 Na 26	0	1.07 s	3+	-0.0053(2)	[23Na]	CFBLS/ β -NMR	2000Ke09	Eur Phys J A8 31 (2000)
	11 Na 27	0	0.30 s	5/2+	-0.0071(3)	[23Na]	CFBLS/ β -NMR	2000Ke09	Eur Phys J A8 31 (2000)
	11 Na 28	0	30.5 ms	1+	+0.389(11)	[23Na]	CFBLS/ β -NMR	2000Ke09	Eur Phys J A8 31 (2000)
	11 Na 29	0	44 ms	3/2+	+0.085(3)	[23Na]	CFBLS/ β -NMR	2000Ke09	Eur Phys J A8 31 (2000)
Magnesium		Calculation of the quadrupole coupling constant of the 3P1 state of the Mg atom							
Reference isotope	12 Mg 22	1247	2 ps	2+	-0.4(4)		CER	2018He12	PL B782 468 (2018)
	12 Mg 23	0	11.3 s	3/2+	0.114(3)	[25Mg]	β -NMR	1999Mb13	HFI 120/121 673 (1999)
	12 Mg 24	1369	1.33 ps	2+	-0.29(3)		CER	1990Gr11	PR C42 R471 (1990)
	12 Mg 25	0	stable	5/2+	+0.199(2)		AB	2008Py02	Mol Phys 106 1965 (2008)
	12 Mg 26	1809	476 fs	2+	-0.21(2)		CER	1991He09	PR C43 2546 (1991)
	12 Mg 29	0	1.3 0 s	3/2+	-0.16(4)	[25Mg]	β -NMR	2019Yo06	HFI 240 67 (2019)
	12 Mg 33	0	90.5 ms	3/2-	+0.13(9)	[25Mg]	β -NMR	2019Yo06	HFI 240 67 (2019)
Aluminium		Calculation of the quadrupole coupling constant of the 3P3/2 state of the Al atom							
Reference isotope	13 Al 25	0	7.18 s	5/2+	0.24(2)	[27Al]	β -NQR	2007Ma94	HFI 180 65 (07)
	13 Al 26	0	7 10 ⁵ y	5+	+0.26(3)	[27Al]	ABLS	1997Le19	JPhys G23 1145 (1997)
	13 Al 27	0	stable	5/2+	+0.1466(10)		AB	2008Py02 1968Ma23	Mol Phys 106 1965 (2008) PRS A305 139 (1968)
	13 Al 28	0	2.24 m	3+	0.172(12)	[27Al]	β -NMR	1978St31	HFI 4 170 (1978)
	13 Al 31	0	644 ms	(5/2+)	0.134(2)	[27Al]	β -NQR	2009De25	PL B678 344 (2009)

	13 Al 32	0	33 ms	1+	0.0255(3)	[27Al]	β -NQR	2018Xu05	PL B782 619 (2018)
	13 Al 33	0	41.7 ms	(5/2+)	0.141(3)	[27Al]	β -NQR	2016He09	PR C94 034312 (2016)
	13 Al 34	47	26 ms	(1+)	0.038(5)	[27Al]	β -NQR	2018Xu05	PL B782 619 (2018)
Silicon	<i>There is no adopted reference efg for Si.</i>								
	<i>A. Efg at Si in Al2O3 estimated from band structure calculations</i>								
	14 Si 27	0	4.15 s	5/2+	0.063(14)	A	β -NQR	1999Mb13	HFI 120/121 673 (1999)
	14 Si 28	1779	0.48 ps	2+	+0.16(3)		CER	1981Sp07	PRep 73 369 (1981)
	14 Si 30	2235	0.22 ps	2+	-0.05(6)		CER	1981Sp07	PRep 73 369 (1981)
Phosphorus	<i>There is no adopted reference efg for P.</i>								
	<i>A. Calculated efg at P site in α-Al2O3</i>								
	15 P 28	0	270 ms	3+	0.137(14)	A	β -NQR	2012Zh36	Chin Phys Lett 29 092102 (2012)
Sulphur	<i>Calculation of the quadrupole coupling constant at S in molecular CS and SiS.</i>								
	<i>A. Efg at S site in FeS2</i>								
	16 S 32	2230	0.17ps	2+	-0.16(2)		CER	1982Ve09	NP A389 185 (1982)
Reference isotope	16 S 33	0	stable	3/2+	-0.694(4)		MA	2018Py01	Mol Phys 116 1328 (2018)
	16 S 34	2128	0.32 ps	2+	+0.04(3)		CER	1981Sp07	PRep 73 369 (1981)
Reference isotope	16 S 35	0	87.4 d	3/2+	-0.483(3)		MA	2018Py01	Mol Phys 116 1328 (2018)
	16 S 43	320	415 ns	7/2-	0.23(3)	A	TDPAD	2012Ch16	PRL 108 162501 (2012)
Chlorine	<i>Calculation of the quadrupole interaction at Cl in the HCl molecule</i>								
Reference isotope	17 Cl 35	0	stable	3/2+	-0.0817(8)		AB	2008Py02	Mol Phys 106 1965 (2008)
	17 Cl 36	0	3.0 10 ⁵ y	2+	-0.178(4)	[35Cl]	MA	1972St38	PR A6 1702 (1972)
Reference isotope	17 Cl 37	0	stable	3/2+	-0.0644(6)		AB	2008Py02	Mol Phys 106 1965 (2008)
Argon	<i>Calculation of the quadrupole coupling constant instates of the the Ar atom</i>								
	18 Ar 35	0	1.78s	3/2+	-0.084(15)	[37Ar]	CFBLS/ β -NMR	1996KI04	NP A607 1 (1996)
	18 Ar 36	1970	0.28 ps	2+	+0.11(6)		CER	1971Na06	PL 34B 389 (1971)
Reference isotope	18 Ar 37	0	35.0 d	3/2+	+0.076(3)		CFBLS/ β -NMR	2018Py01	Mol Phys 116 1328 (2018)

Reference isotope	18 Ar 39	0	269 y	7/2-	-0.116(2)		CLS	2018Py01	Mol Phys 116 1328 (2018)
	18 Ar 40	1461	1.12 ps	2+	+0.01(4)		CER	1971Na05	PRL 24 903 (1970)
	18 Ar 41	0	1.82 h	7/2-	-0.042(4)	[37Ar]	CFBLS	2008BI01	NP A799 30 (2008)
	18 Ar 43	0	5.37 m	5/2-	+0.142(14)	[37Ar]	CFBLS	2008BI01	NP A799 30 (2008)
Potassium <i>Calculation of the quadrupole coupling constant at K in KF, KCl and KBr.</i>									
	19 K 37	0	1.23 s	3/2+	+0.109(4)	[39K]	β -NQR	2008MI07	PL B662 389 (2008)
Reference isotope	19 K 39	0	stable	3/2+	+0.603(6)		AB	2018Py01 1998Ke05	Mol Phys 116 1328 (2018) CPL 292 403 (1998)
Reference isotope	19 K 40	0	1.25x10 ⁹ y	4-	-0.750(8)		AB	2018Py01 1998Ke05	Mol Phys 116 1328 (2018) CPL 292 403 (1998)
Reference isotope	19 K 41	0	stable	3/2+	+0.734(7)		AB	2018Py01 1998Ke05	Mol Phys 116 1328 (2018) CPL 292 403 (1998)
Calcium <i>Calculation of the quadrupole coupling constant of the 1D2 state of the Ca atom</i>									
	20Ca37	0	181 ms	(3/2+)	-0.15(11)		CLS	2019KL06	PR C99 061301 (2019)
	20 Ca 39	0	0.86 s	3/2+	+0.038(3)		CLS	2019KL06	PR C99 061301 (2019)
Reference isotope	20 Ca 41	0	1.0x10 ⁵ y	7/2-	-0.0665(18)		AB	2008Py02	Mol Phys 106 1965 (2008)
	20 Ca 42	1525 2424	0.83 ps 140 fs	2+ 2+	-0.19(8) -0.42(12)		CER CER	1973To07 2018Ha10	NP A204 574 (1973) PR C97 024326 (2018)
Reference isotope	20 Ca 43	0	stable	7/2-	-0.0408(8)		AB	2008Py02	Mol Phys 106 1965 (2008)
	20 Ca 44	1157	2.7 ps	2+	-0.14(7)		CER	1973To07	NP A204 574 (1973)
	20 Ca 45	0	162.6 d	7/2-	+0.038(12)	[41Ca]	ABLFS	1983Ar25	ZP A314 303 (1983)
	20 Ca 47	0	4.54 d	7/2-	+0.084(6)	[43Ca]	CLS	2015Ru02	PR C91 041304 Rapid Report
	20 Ca 49	0	8.72 m	3/2-	-0.036(3)	[43Ca]	CLS	2015Ru02	PR C91 041304 Rapid Report
	20 Ca 51	0	10.0 s	3/2-	+0.036(12)	[43Ca]	CLS	2015Ru02	PR C91 041304 Rapid Report
Scandium <i>Calculation of the quadrupole coupling constants in ScF, ScCl and ScBr molecules</i>									
	21 Sc 41	0	0.60 s	7/2-	-0.145(3)	[45Sc]	β -NQR	2002Mi37	ZNat 57a 595 (2002)
	21 Sc 43	0 3124	3.89 h 472 ns	7/2- 19/2-	-0.27(5) 0.199(14)	[45Sc] [45Sc]	CLS TDPAD	2011Av01 1981Da06	J Phys G38 025104 (2011) PR C23 1612 (1981)

Reference isotope	21 Sc 44	0 68 271	3.97h 155 ns 58.6 h	2+ 1- 6+	+0.10(5) 0.21(2) -0.19(2)	[45Sc] [45Sc] [45Sc]	CLS TDPAC CLS	2011Av01 1973Ha61 2011Av01	J Phys G38 025104 (2011) JCP 58 3339 (1973) J Phys G38 025104 (2011)
	21 Sc 45	0 12.4	stable 318 ms	7/2- 3/2+	-0.220(2) +0.28(5)	[45Sc]	MS CLS	2008Py02 2011Av01	Mol Phys 106 1965 (2008) J Phys G38 025104 (2011)
	21 Sc 46	0	83.79 d	4+	+0.119(6)	[45Sc]	AB	1962Pe21	PR 128 1740 (1962)
	21 Sc 47	0	3.35 d	7/2-	-0.22(3)	[45Sc]	AB	1966Co13	PR 141 1106 (1966)
Titanium	Calculation of the quadrupole coupling constants in states of the Ti+ ion								
Reference isotope	22 Ti 43	3066	556 ns	19/2-	0.33(8)	[47Ti]	TDPAD	1981Da06	PR C23 1612 (1981)
	22 Ti 45	0	184.8 m	7/2-	0.015(15)	[47Ti][49Ti]	AB	1966Co19	PR 148 1157 (1966)
	22 Ti 46	889	5.32 ps	2+	-0.21(6)		CER	1975To06	NP A250 381 (1975)
	22 Ti 47	0	stable	5/2-	+0.302(10)		AB	2008Py02	Mol Phys 106 1965 (2008)
	22 Ti 48	984	4.04 ps	2+	-0.177(8)		ES	1972Li12	PL B38 475 (1972)
Reference isotope	22 Ti 49	0	stable	7/2-	+0.247(11)		AB	2008Py02	Mol Phys 106 1965 (2008)
	22 Ti 50	1554	1.12 ps	2+	+0.08(16)		CER	1975To06	NP A250 381 (1975)
Vanadium	Calculation of the quadrupole coupling constants in states of the V atom								
A. Calculated efg in 3d/4s excited states of the V atom									
Reference isotope	23 V 50	0	1.4 10 ¹⁷ y	6+	+0.21(4)		ABLDF	2018Py01 1979Er04	Mol Phys 116 1328 (2018) PL B85 319 (1979)
	23 V 51	0	stable	7/2-	-0.052(10)	A	LRFS	2018Py01 1989Un01	Mol Phys 116 1328 (2018) ZP D11 259 (1989)
Chromium	Calculation of the quadrupole coupling constants in states of the Cr atom								
Reference isotope	24 Cr 50	783	9.67 ps	2+	-0.36(7)		CER	1975To06	NP A250 381 (1975)
	24 Cr 52	1434	0.783 ps	2+	-0.08(2)		ES	1989Ra17	JPSJ 34 387 (1973)
	24 Cr 53	0	stable	3/2-	-0.15(5)		AB	2018Py01	Mol Phys 116 1328 (2018)
	24 Cr 54	835	8.0 ps	2+	-0.21(8)		CER	1975To06	NP A250 381 (1975)
Manganese	Calculation of the quadrupole coupling constant for the 6D states of the Mn atom								
	25 Mn 50	229	1.75 m	5+	+0.83(12)	[Mn55]	TLS	2010Ch15	PL B690 346 (2010)
	25 Mn 51	0	46.2 m	5/2-	0.41(8)	[Mn55]	AB	1971Jo10	NP A166 306 (1971)

<i>Reference isotope</i>	25 Mn 52	0	5.80 d	6+	+0.50(7)	[Mn55]	NMR/ON	1970Ni11	Phca 50 259 (1970)
	25 Mn 53	0	3.7 10 ⁶ y	7/2-	+0.17(3)	[Mn55]	TLS	2010Ch15	PL B690 346 (2010)
	25 Mn 54	0	312 d	3+	+0.37(3)	[Mn55]	TLS	2010Ch15	PL B690 346 (2010)
	25 Mn 55	0	stable	5/2-	+0.330(10)		AB	2018Py01	Mol Phys 116 1328 (2018)
	25 Mn 56	0	2.58 h	3+	+0.24(8)	[Mn55]	CLS	2015He28	PR C92 044311 (2015)
	25 Mn 57	0	85.4 s	5/2-	+0.37(3)	[Mn55]	CLS	2016Ba44	PL B760 387 (2016)
	25 Mn 58	0	3.0 s	1+	+0.03(8)	[Mn55]	CLS	2015He28	PR C92 044311 (2015)
		72	65.4 s	4+	+0.47(9)	[Mn55]	CLS	2015He28	PR C92 044311 (2015)
	25 Mn 59	0	4.6 s	5/2-	+0.34(3)	[Mn55]	CLS	2016Ba44	PL B760 387 (2016)
	25 Mn 60	0	0.28 s	1+	+0.1(2)	[Mn55]	CLS	2015He28	PR C92 044311 (2015)
		272	1.77 s	4+	+0.33(10)	[Mn55]	CLS	2015He28	PR C92 044311 (2015)
	25 Mn 61	0	0.71 s	5/2-	+0.36(3)	[Mn55]	CLS	2016Ba44	PL B760 387 (2016)
	25 Mn 62	0	92 ms	1+	+0.1(2)	[Mn55]	CLS	2015He28	PR C92 044311 (2015)
		0 + x	671 ms	4+	+0.59(13)	[Mn55]	CLS	2015He28	PR C92 044311 (2015)
	25 Mn 63	0	0.28 s	5/2-	+0.48(4)	[Mn55]	CLS	2016Ba44	PL B760 387 (2016)
	25 Mn 84	0	90 ms	1+	+0.21(11)	[Mn55]	CLS	2015He28	PR C92 044311 (2015)

Iron	<i>Efg calculations in many Fe compounds</i>								
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<i>Reference isotope</i>	26 Fe 54	1408	0.76 ps	2+	-0.05(14)	[57Fe 14 keV]	CER	1981Le02	PR C23 244 (1981)
		6527	364 ns	10+	+0.30(4)		TDPAD/TF	1984Ha07	NP A414 316 (1984)
	26 Fe 56	847	6.1 ps	2+	-0.23(3)		CER	1971Th14	PR C4 1699 (1971)
	26 Fe 57	14	98 ns	3/2-	+0.160(8)		ME	2018Py01 1995Du17	Mol Phys 116 1328 (2018) PRL 75 3545 (1995)
	26 Fe 58	811	6.5 ps	2+	-0.27(5)		CER	1981Le02	PR C23 244 (1981)
	26 Fe 61	862	238 ns	(9/2+)	0.44(6)	[57Fe 14 keV]	TDPAD	2007Ve05	PR C75 051302 (2007)

Cobalt	<i>Calculation of the quadrupole coupling constants in states of the Co atom</i>								
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27 Co 56	0	77.2 d	4+	+0.25(9)	[59Co]	MAPON	1988Ba87	PR B37 4911 (1988)
27 Co 57	0	272 d	7/2-	+0.54(10)	[59Co]	NMR/ON	1972Ni01	Phca 57 1 (1972)
27 Co 58	0	70.8 d	2+	+0.23(3)	[59Co]	NMR/ON	1972Ni01	Phca 57 1 (1972)

Reference isotope	27 Co 59	0	stable	7/2-	+0.42(3)		AB	2018Py01	Mol Phys 116 1328 (2018)
	27 Co 60	0	5.271 y	5+	+0.46(6)	[59Co]	NMR/ON	1972Ni01	Phca 57 1 (1972)
Nickel <i>Calculation of the quadrupole coupling constants in states of the Ni atom</i>									
	28 Ni 58	1454	0.65 ps	2+	-0.10(6)		CER	1974Le13	NP A223 563 (1974)
	28 Ni 60	1332	0.74 ps	2+	-0.10(2)		ES	1972Li12	PL 38B 475 (1972)
Reference isotope	28 Ni 61	0	stable	3/2-	+0.162(15)		AB	2018Py01	Mol Phys 116 1328 (2018)
		67	5.34 ns	5/2-	-0.20(3)	[61Ni]	ME	1968Ch10 1971Go31	PR 170 136 (1968) ZNat 26a 1931 (1971)
	28 Ni 62	1173	1.45 ps	2+	+0.05(12)		CER	1974Le13	NP A223 563 (1974)
	28 Ni 64	1346	1.09 ps	2+	+0.4(2)		CER	1971ChZK	BAPS 16 625 (1971)
Copper <i>Muonic atom X-ray hyperfine structure</i>									
	29 Cu 58	0	3.2 s	1+	-0.16(3)	[65Cu]	CLS/R	2011Vi03	PL B703 34 (2011)
	29 Cu 59	0	81.5 s	3/2-	-0.20(2)	[65Cu]	CLS/R	2011Vi03	PL B703 34 (2011)
	29 Cu 60	0	23.7 m	2+	+0.121(13)	[65Cu]	CLS/R	2011Vi03	PL B703 34 (2011)
	29 Cu 61	0	3.34 h	3/2-	-0.221(10)	[65Cu]	CLS/R	2011Vi03	PL B703 34 (2011)
	29 Cu 62	0	9.67 m	1+	-0.022(4)	[65Cu]	CLS/R	2011Vi03	PL B703 34 (2011)
Reference isotope	29 Cu 63	0	stable	3/2-	-0.220(15)		Mu-X	2018Py01 1982Ef01	Mol Phys 116 1328 (2018) ZP A309 77 (1982)
	29 Cu 64	0	12.7 h	1+	+0.075(9)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
Reference isotope	29 Cu 65	0	stable	3/2-	-0.204(14)		Mu-X	2018Py01 1982Ef01	Mol Phys 116 1328 (2018) ZP A309 77 (1982)
	29 Cu 66	0	5.1 m	1+	+0.059(14)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
		1154	0.60 μ s	6-	(+)0.195(13)	[63Cu,65Cu]	TDPAD	2011Lo01	PL B694 316 (2011)
	29 Cu 67	0	61.8 h	3/2-	-0.182(8)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
	29 Cu 68	0	30.9 s	1+	-0.086(14)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
		84	7.84 ns	2+	0.115(8)	[65Cu]	TDPAC/R	2016Fe08	Eur Phys Lett 115 62002 (2016)
		721	3.75 m	6-	-0.46(2)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
	29 Cu 69	0	2.85 m	3/2-	-0.154(17)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
	29 Cu 70	0	44.5 s	6-	-0.298(15)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)

	101	33 s	3-	-0.14(4)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
	242	6.6 s	1+	-0.12(3)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
29 Cu 71	0	19.4 s	3/2-	-0.200(17)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
29 Cu 72	0	6.62 s	2-	+0.08(2)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
29 Cu 73	0	4.2 s	3/2-	-0.210(10)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
29 Cu 74	0	1.63 s	2-	+0.27(3)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
29 Cu 75	0	1.22 s	5/2-	-0.281(17)	[65Cu]	CLS/R	2010Vi07	PR C82 064311 (2010)
29 Cu 76	0	641 ms	3-	+0.36(2)	[65Cu]	CLS/R	2017De30	PR C96 041302(R) (2017)
29 Cu 77	0	470 ms	5/2+	-0.27(3)	[65Cu]	CLS/R	2017De30	PR C96 041302(R) (2017)
29 Cu 78	0	335 ms	(6-)	+0.02(10)	[65Cu]	CLS/R	2017De30	PR C96 041302(R) (2017)

Zinc	Calculation of the quadrupole coupling constants in states of the Zn atom							2018Bi07	PR A97 062505 (2018)
	30 Zn 63	0	38.4 m	3/2-	+0.20(2)	[67Zn]	CLS	1969La05	PR 177 1606 (1969)
	30 Zn 64	992	1.94 ps	2+	-0.14(2)		ES	1981Ko06 1976Ne06	JPhys G7 L63 (1981) NP A263 249 (1976)
	30 Zn 65	0	244.1 d	5/2-	-0.019(2)	[67Zn]	OD	1964By01	PR 134 A47 (1964)
	30 Zn 66	1039 4074	1.68 ps 30 ps	2+ 6-	+0.24(8) -0.081(13)		CER ES	2003Ko51 1981Ko06 1976Ne06	Eur Phys J A18 87 (2003) JPhys G7 L63 (1981) NP A263 249 (1976)
Reference isotope	30 Zn 67	0	stable	5/2-	+0.122(10)		CLS	2017Wr01 2018Bi07	PL B771 385 (2017) PR A97 062505 (2018)
		604	333 ns	9/2+	+0.44(4)	[67Zn]	NQR	1976Ch37 1979Ka44	ZP B24 177 (1976) Sol St Comm 29 375 (1979)
	30 Zn 68	1077	1.61 ps	2+	-0.106(16)		ES	1981Ko06 1976Ne06	JPhys G7 L63 (1981) NP A263 249 (1976)
	30 Zn 69	439	13.7 h	9/2+	-0.39(3)	[67Zn]	CLS	1983Oe01	ZP A310 233 (1983)
	30 Zn 70	885	3.7 ps	2+	-0.24(3)		ES	1981Ko06 1976Ne06	JPhys G7 L63 (1981) NP A263 249 (1976)
	30 Zn 71	158	3.96 h	9/2+	-0.26(3)	[67Zn]	CLS	2017Wr01	PL B771 385 (2017)
	30 Zn 73	195	13.0 ms	5/2+	+0.43(4)	[67Zn]	CLS	2017Wr01	PL B771 385 (2017)
	30 Zn 75	0	10.2 s	7/2+	+0.16(2)	[67Zn]	CLS	2017Wr01	PL B771 385 (2017)

	30 Zn 77	0	2.08 s	7/2+	+0.48(4)	[67Zn]	CLS	2017Wr01	PL B771 385 (2017)
	30 Zn 79	0	0.75 s	9/2+	+0.40(4)	[67Zn]	CLS	2017Wr01	PL B771 385 (2017)
Gallium	<i>Calculation of the quadrupole coupling constants in GaF, GaCl and GaBr molecules</i>								
	31 Ga 63	0	32.4 s	3/2-	+0.212(14)	[69Ga]	CLS	2012Pr11	PR C86 034329 (2012)
	31 Ga 65	0	15.2 m	3/2-	+0.210(15)	[71Ga]	CLS	2017Fa09	PR C96 044324 (2017)
	31 Ga 66	1464	57 ns	7-	+0.78(4)	[69Ga][71Ga]	TDPAD	1985Ra33	HFI 26 855 (1985)
	31 Ga 67	0	78.3 h	3/2-	+0.197(2)	[69Ga][71Ga]	AB	1968Eh02 2001Py02	PR 176 25 (1968) Mol Phys 99 1617 (2001)
	31 Ga 68	0 1230	67.7 m 62 ns	1+ 7-	-0.0277(14) +0.72(2)	[69Ga][71Ga] [69Ga][71Ga]	AB TDPAD	1972St38 1985Ra33	PR A6 1702 (1972) HFI 26 855 (1985)
<i>Reference isotope</i>	31 Ga 69	0	stable	3/2-	+0.171(2)		MS	2018Py01	Mol Phys 116 1328 (2018)
	31 Ga 70	0	21.1 m	1+	+0.105(7)	[69Ga]	CLS	2012Pr11	PR C86 034329 (2012)
<i>Reference isotope</i>	31 Ga 71	0	stable	3/2-	+0.107(1)		MS	2018Py01	Mol Phys 116 1328 (2018)
	31 Ga 72	0	14.1 h	3-	+0.530(6)	[69Ga][71Ga]	AB	1968Eh02 2001Py02	PR 176 25 (1968) Mol Phys 99 1617 (2001)
	31 Ga 73	0	4.86 h	3/2-	+0.209(2)	[71Ga]	CLS	2010Ch16	PRL 104 252502 (2010)
	31 Ga 74	0	8.12 m	3- or 4-	+0.55(4) or +0.60(4)	[71Ga]	LRS	2011Ma45	PR C84 024303 (2011)
	31 Ga 75	0	126 s	3/2-	-0.285(17)	[71Ga]	CLS	2010Ch16	PRL 104 252502 (2010)
	31 Ga 76	0	32.6 s	(2+)	+0.33(2)	[71Ga]	LRS	2011Ma45	PR C84 024303 (2011)
	31 Ga 77	0	13.2 s	3/2-	-0.208(13)	[71Ga]	CLS	2010Ch16	PRL 104 252502 (2010)
	31 Ga 78	0	5.1 s	(2+)	+0.33(2)	[71Ga]	LRS	2011Ma45	PR C84 024303 (2011)
	31 Ga 79	0	2.85 s	3/2-	+0.158(10)	[71Ga]	CLS	2010Ch16	PRL 104 252502 (2010)
	31 Ga 80	0 22.4	1.9 s 1.3 s	(3-) (6-)	+0.38(2) +0.48(3)	[71Ga] [71Ga]	CLS CLS	2010Ch50 2010Ch50	PR C82 051302(R) (2010) PR C82 051302(R) (2010)
	31 Ga 81	0	1.22 s	5/2-	-0.048(8)	[71Ga]	CLS	2010Ch16	PRL 104 252502 (2010)
Germanium	<i>Calculation of the quadrupole coupling constants in GeO, GeS molecules</i>								
	<i>A. Efg of Ge in Zn single crystal</i>								
	32 Ge 67	752	146 ns	9/2+	0.92(9)	[73Ge]	TDPAD	1993Co17 1981Vi05	HFI 80 1321 (1993) HFI 10 1243 (1981)

	32 Ge 69	0 398	39.0 h 2.8 μ s	5/2- 9/2+	+0.114(7) 0.75(8)	[73Ge] [73Ge]	CLS TDPAD	2020Ka58 1993Co17 1981Vi05	PR C102 054331 (2020) HFI 80 1321 (1993) HFI 10 1243 (1981)
	32 Ge 70	1039	1.30 ps	2+	+0.03(6)		CER	1980Le16 2000To12	PR C22 1530 (1980) Eur Phys J A9 353 (2000)
	32 Ge 71	175 199	79 ns 20.4 ms	5/2+ 9/2+	0.18(4) 0.34(5)	A	TDPAD QIR	1993Co17 1981Vi05 1975Ri03 1976Br41	HFI 80 1321 (1993) HFI 10 1243 (1981) PS 11 228 (1975) HFI 2 265 (1976)
	32 Ge 72	834	3.35 ps	2+	-0.13(6)		CER	1980Le16 2000To12	PR C22 1530 (1980) Eur Phys J A9 353 (2000)
Reference isotope	32 Ge 73	0	stable	9/2+	-0.196(1)		MS	2018Py01 1999Ke17	Mol Phys 116 1328 (2018) Mol Phys 96 275 (1999)
		13	2.92 μ s	5/2+	0.70(8)	A	TDPAC	1993Co17 1981Vi05	HFI 80 1321 (1993) HFI 10 1243 (1981)
	32 Ge 74	596	12.4 ps	2+	-0.19(2)		CER	2000To12	Eur Phys J A9 353 (2000)
		1204	5.4 ps	2+	-0.26(6)		CER	2000To12	Eur Phys J A9 353 (2000)
	32 Ge 76	563	18.2 ps	2+	-0.19(6)		CER	1980Le16 2000To12	PR C22 1530 (1980) Eur Phys J A9 353 (2000)
Arsenic		Calculation of the quadrupole coupling constant in AsP.						2010De44	Chem Phys Lett 498 10 (2010)
	33 As 70	0	53 m	4+	+0.09(2)	[75As]	AB	1980Ho02	ZP A294 1 (1980)
	33 As 71	0	65.3 h	5/2-	-0.021(6)	[75As]	NO/S	1988Wh03	HFI 43 205 (1988)
	33 As 72	0	26 h	2-	-0.08(2)	[75As]	AB	1980Ho02	ZP A294 1 (1980)
	33 As 73	66	5.0 ns	5/2-	+0.353(12)	[75As]	TDPAC	1992Sc21	ZP A343 279 (1992)
Reference isotope	33 As 75	0	stable	3/2-	+0.311(2)		MS	2018Py01 2010De44	Mol Phys 116 1328 (2018) Chem Phys Lett 498 10 (2010)
Selenium		Calculation of the quadrupole coupling constant in Se metal							
	34 Se 70	945	1.0 ps	2+	-0.6(3)		CER	2018He14	PRL 121 082502 (2018)
	34 Se 72	862	2.8 ps	2+	-0.6(3)		CER	2018He14	PRL 121 082502 (2018)
	34 Se 74	635	7.08 ps	2+	-0.36(7)		CER	1978Le22	PR C18 2801 (1978)
	34 Se 75	0	119.8 d	5/2+	1.1(2)	[77Se]	MA	1955Aa06	PR 98 1224 (1955)
	34 Se 76	559	12.3 ps	2+	-0.35(4)		CER	2019He07	PR C99 054313 (2019)

		1216	3.4 ps	2+	+0.19(4)		CER	2019He07	PR C99 054313 (2019)
		1331	1.52 ps	4+	-0.29(4)		CER	2019He07	PR C99 054313 (2019)
Reference isotope	34 Se 77	250	9.6 ns	5/2-	+0.76(5)		TDPAC	2018Py01 1983Un02	Mol Phys 116 1328 (2018) HFI 14 119 (1983)
	34 Se 78	614	9.8 ps	2+	-0.26(9)		CER	1977Le11	NP A284 123 (1977)
	34 Se 79	0	3.3 10 ⁵ y	7/2+	+0.8(2)	[77Se]	MA	1989Ra17	ADNDT 42 189 (1989) OSpk 12 163 (1962)
	34 Se 80	666	8.5 ps	2+	-0.31(7)		CER	1977Le11	NP A284 123 (1977)
	34 Se 82	655	13.1 ps	2+	-0.22(7)		CER	1977Le11	NP A284 123 (1977)
Bromine <i>Calculation of the quadrupole coupling constants in states of the Br atom.</i>									
	35 Br 76	0	16.2 h	1-	+0.251(4)	[79Br]	AB	1960Li11	PR 119 1053 (1960)
	35 Br 77	0	57 h	3/2-	+0.50(2)	[79Br]	MAPON	1998Se09	PRL 80 5289 (1998)
Reference isotope	35 Br 79	0	stable	3/2-	+0.3087(2)		AB	2018Py01 2013Ch52	Mol Phys 116 1328 (2018) J Phys Chem A117 12616 (2013)
	35 Br 80	0	17.7 m	1+	+0.182(3)	[79Br]	AB	1964Wh05	PR 136 B584 (1964)
		37	7.4 ns	2-	0.162(6)	[79Br]	AB	1978Ta24	HP Ac 51 755 (1978)
		86	4.42 h	5-	+0.700(10)	[79Br]	AB	1964Wh05	PR 136 B584 (1964)
Reference isotope	35 Br 81	0	stable	3/2-	+0.2579(2)	[79Br]	AB/MS	2018Py01 2013Ch52	Mol Phys 116 1328 (2018) J Phys Chem A117 12616 (2013)
	35 Br 82	0	35.3 h	5-	+0.697(10)	[79Br]	AB	1959Ga12	PR 116 393 (1959)
Krypton <i>Calculation of the quadrupole coupling constants in KrH+</i>									
	36 Kr 75	0	4.6 m	5/2+	+1.137(13)	[83Kr]	CFBLS	1995Ke04	NP A586 219 (1995)
	36 Kr 77	0	74.4 m	5/2+	+0.948(10)	[83Kr]	CFBLS	1995Ke04	NP A586 219 (1995)
	36 Kr 79	130	50 s	7/2+	+0.404(5)	[83Kr]	CFBLS	1995Ke04	NP A586 219 (1995)
		147	78.7 ns	5/2-	+0.45(3)	[83Kr]	TDPAD	1978HaXP	ARHMI 50 (1977)
	36 Kr 81	0	2.3 10 ⁵ y	7/2+	+0.644(4)	[83Kr]	LRFS	1993Ca41	PR A47 1148 (1993)
Reference isotope	36 Kr 83	0	stable	9/2+	+0.259(1)		MS	2018Py01	Mol Phys 116 1328 (2018)
		9	157 ns	7/2+	+0.507(3)	[83Kr]	ME	1977Ho33	JCP 66 2627 (1977)
	36 Kr 84	3236	1.83 μs	8+	+0.36(4)	[83Kr]	LEMS	2006Sc22	PR C74 034309 (2006)
	36 Kr 85	0	10.74 y	9/2+	+0.443(3)	[83Kr]	LRFS	1993Ca41	PR A47 1148 (1993)

36 Kr 87	0	76.3 m	5/2+	-0.300(3)	[83Kr]	CFBLS	1995Ke04	NP A586 219 (1995)
36 Kr 89	0	3.15 m	3/2+	+0.166(2)	[83Kr]	CFBLS	1995Ke04	NP A586 219 (1995)
36 Kr 91	0	8.57 s	5/2+	+0.303(6)	[83Kr]	CFBLS	1995Ke04	NP A586 219 (1995)
36 Kr 94	666	8.7 ps	2+	-0.5(3)		CER	2012 AI03	PRL 108 062701 (2012)

Rubidium	Calculation of the quadrupole coupling constants in RbF								
	37 Rb 75	0	19.0 s	3/2(-)	+0.81(4)	[85Rb]	CFBLS	2015Pr05	Eur. Phys. J. A 51 23 (2015)
	37 Rb 76	0	36.5 s	1(-)	+0.41(9)	[85Rb]	CFBLS	2015Pr05	Eur. Phys. J. A 51 23 (2015)
	37 Rb 77	0	3.8 m	3/2-	+0.84(17)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 78	103	5.7 m	4-	+1.07(3)	[85Rb]	CFBLS	2015Pr05	Eur. Phys. J. A 51 23 (2015)
	37 Rb 79	0	23 m	5/2+	-0.12(4)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 80	0	33.4 s	1+	+0.42(8)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 81	0	4.57 h	3/2-	+0.48(10)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
		86	30.5 m	9/2+	-0.90(19)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 82	0	1.26 m	1+	+0.23(10)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
		69	6.47 h	5-	+1.22(27)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 83	0	86.2 d	5/2-	+0.24(5)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 84	0	32.8 d	2-	-0.02(4)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
		464	20.3 m	6-	+0.70(36)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
Reference isotope	37 Rb 85	0	stable	5/2-	+0.276(1)		MS	2018Py01	Mol Phys 116 1328 (2018)
		514	1.02 μs	9/2+	-0.9(3)	[85Rb]	OPD	1991Ma21	PRL 66 1681 (1991)
	37 Rb 86	0	18.64 d	2-	+0.23(6)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
		556	1.02 m	(6-)	+0.45(14)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
Reference isotope	37 Rb 87	0	5.0 10 ¹⁰ y	3/2-	+0.1335(5)		MS	2018Py01	Mol Phys 116 1328 (2018)
	37 Rb 88	0	17.8 m	2-	-0.01(11)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 89	0	15.3 m	3/2-	+0.17(3)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 90	107	4.26 m	3-	+0.25(7)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 91	0	58.2 s	3/2(-)	+0.19(5)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 93	0	5.84 s	5/2-	+0.21(6)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)

	37 Rb 94	0	2.70 s	3(-)	+0.20(7)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 95	0	0.38 s	5/2-	+0.26(9)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 96	0	0.20 s	2+	+0.30(9)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 97	0	0.17 s	3/2-	+0.70(15)	[85Rb]	ABLS	1981Th04	PR C23 2720 (1981)
	37 Rb 98	~ 270	96 ms	(3+)	+1.73(4)	[85Rb]	CFBLS	2015Pr05	Eur. Phys. J. A 51 23 (2015)
Strontium	<i>Calculation of the quadrupole coupling constants in the 4d 2D5/2 and 5P3/2 states of the Sr+ ion</i>								
	38 Sr 77	0	9 s	5/2+	+1.27(5)	[87Sr]	CFBLS	1992Li11	PR C46 797 (1992)
	38 Sr 79	0	2.25 m	(3/2-)	+0.661(6)	[87Sr]	CFBLS	1990Bu12	PR C41 2883 (1990)
	38 Sr 83	0	32.4 h	7/2+	+0.708(11)	[87Sr]	CFBLS	1990Bu12	PR C41 2883 (1990)
	38 Sr 85	0	64.8 d	9/2+	+0.263(14)	[87Sr]	CFBLS	1990Bu12	PR C41 2883 (1990)
<i>Reference isotope</i>	38 Sr 87	0	stable	9/2+	+0.305(2)		AB	2008Py02 2006Sa21	Mol Phys 106 1965 (2008) PR A73 062501 (2006)
	38 Sr 89	0	50.6 d	5/2+	-0.253(8)	[87Sr]	CFBLS	1990Bu12	PR C41 2883 (1990)
	38 Sr 91	0	9.65 h	5/2+	+0.042(10)	[87Sr]	CFBLS	1990Bu12	PR C41 2883 (1990)
	38 Sr 93	0	7.4 m	2+	+0.240(10)	[87Sr]	CFBLS	1990Bu12	PR C41 2883 (1990)
	38 Sr 96	815	4.8 ps	2+	-0.2(3)		CER	2016CI01	PRL 116 022701 (2016)
	38 Sr 98	144	2.8 ns	4+	-0.5(2)		CER	2016CI01	PRL 117 099902 (2016)
		433	82 ps	6+	-2.1(2)		CER	2016CI01	PRL 116 022701 (2016)
		867	7.9 ps	2+	-1.3(2)		CER	2016CI01	PRL 117 099902 (2016)
		871	8.6 ps	8+	+0.03(20)		CER	2016CI01	PRL 116 022701 (2016)
		1432	3 ps	5/2+	-0.9(9)		CER	2016CI01	PRL 117 099902 (2016)
	38 Sr 99	0	0.269 s	3/2+	+0.76(4)	[87Sr]	CFBLS	1991Li05	PL B256 141 (1991)
Yttrium	<i>Calculation of the quadrupole coupling constants in the 4d5s2 2D states of the Y atom</i>								
	39 Y 87	381	13.4 h	9/2+	-0.50(6)	[90Y]	CLS	2007Ch07	PL B645 133 (2007)
	39 Y 88	0	106 d	4-	+0.16(3)	[90Y]	CLS	2007Ch07	PL B645 133 (2007)
		675	14 ms	8+	+0.06(6)	[90Y]	CLS	2007Ch07	PL B645 133 (2007)

Reference isotope	39 Y 89	909	15.7 s	9/2+	-0.43(6)	[90Y]	CLS	2007Ch07	PL B645 133 (2007)
	39 Y 90	0	64.0 h	2-	-0.125(11)		AB	2018Py01	Mol Phys 116 1328 (2018)
		682	3.19 h	7+	-0.65(8)	[90Y]	CLS	1998Bi20 2007Ch07	PR A58 4401 (1998) PL B645 133 (2007)
	39 Y 92	0	3.54 h	2-	0.00(2)	[90Y]	CLS	2007Ch07	PL B645 133 (2007)
	39 Y 93	758	0.82 s	9/2+	-0.64(8)	[90Y]	CLS	2007Ch07	PL B645 133 (2007)
	39 Y 94	0	18.7 m	2-	-0.03(3)	[90Y]	CLS	2007Ch07	PL B645 133 (2007)
	39 Y 96	1140	9.6 s	8+	-0.98(11)	[90Y]	CLS	2007Ch07	PL B645 133 (2007)
	39 Y 97	668 3522	1.17 s 142 ms	9/2+ (27/2)	-0.76(8) -1.21(14)	[90Y] [90Y]	CLS CLS	2007Ch07 2007Bi14	PL B645 133 (2007) PL B645 330 (2007)
	39 Y 98	410	2.0 s	4 or 5	+1.7(2) or +1.8(2)	[90Y]	CLS	2007Bi14	PL B645 330 (2007)
	39 Y 99	0	1.48 s	5/2+	+1.55(17)	[90Y]	CLS	2007Bi14	PL B645 330 (2007)
	39 Y 100	(143)	0.94 s	4	+1.85(20)	[90Y]	CLS	2007Bi14 2010Ba31	PL B645 330 (2007) J Phys G37 105103 (2010)
	39 Y 101	0	0.45 s	5/2+	+1.53(17)	[90Y]	CLS	2007Bi14	PL B645 330 (2007)
	39 Y 102	0 + x	0.3 s	2 or 3	+1.17(13) or +1.36(16)	[90Y]	CLS	2007Bi14	PL B645 330 (2007)

Zirconium *Calculation of the quadrupole coupling constants in the ZrO and ZrS molecules*

Reference isotope	40 Zr 87	0	1.68 h	9/2+	+0.42(5)	[91Zr]	CLS	2003Th03	J Phys G29 2247 (2003)
	40 Zr 88	2889	1.32 μ s	8+	+0.44(3)	[91Zr]	TDPAD/TFLD	1985Ra09 1986Be06	PRL 54 2592 (1985) PR C33 1517 (1986)
	40 Zr 89	0	78.4 h	9/2+	+0.28(10)	[91Zr]	CLS	2003Th03	J Phys G29 2247 (2003)
	40 Zr 90	3589	131 ns	8+	-0.44(3)	[91Zr]	TDPAD/TFLD	1985Ra09 1986Be06	PRL 54 2592 (1985) PR C33 1517 (1986)
	40 Zr 91	0	stable	5/2+	-0.176(3)		MS	2018Py01	Mol Phys 116 1328 (2018)
		3167	4.4 μ s	21/2+	0.71(4)	[91Zr]	TDPAD	2000Ke03 1985Ra09	CPL 318 222 (2000) PRL 54 2592 (1985)
	40 Zr 95	0	64.0 d	5/2+	+0.22(2)	[5- 90mZr calc]	MAPON	1998Se01	PRL 80 924 (1998)
	40 Zr 101	0	2.3 s	3/2+	+0.81(6)	[91Zr]	CLS	2002Ca37	PRL 89 082501 (2002)

Niobium *Muonic atom X-ray hyperfine structure*

41 Nb 90	0	14.6 h	8+	+0.01(4)	[93Nb]	CLS	2009Ch25	PRL 102 222501 (2009)
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		125	18.8 s	4-	-0.26(4)	[93Nb]	CLS	2009Ch25	PRL 102 222501 (2009)
	41 Nb 91	0	680 y	9/2+	-0.25(3)	[93Nb]	CLS	2009Ch25	PRL 102 222501 (2009)
	41 Nb 92	0	3.5 10 ⁷ y	7+	-0.35(3)	[93Nb]	CLS	2009Ch25	PRL 102 222501 (2009)
Reference isotope	41 Nb 93	0	stable	9/2+	-0.32(2)		Mu-X	2018Py01 1973Po15	Mol Phys 116 1328 (2018) NP A217 573 (1973)
	41 Nb 99	0	15 s	9/2+	-0.41(14)	[93Nb]	CLS	2009Ch25	PRL 102 222501 (2009)
	41 Nb 101	0	7.1 s	5/2+	+1.05(7)	[93Nb]	CLS	2009Ch25	PRL 102 222501 (2009)
	41 Nb 103	0	1.5 s	5/2+	+1.08(9)	[93Nb]	CLS	2009Ch25	PRL 102 222501 (2009)

Molybdenum *Estimation of the quadrupole coupling constant in states of the Mo atom*
A. Normalised to Q of 92Mo 2760 keV state estimated from B(E2)

	42 Mo 90	2875	1.1 μs	8+	0.61(3)	A	TDPAD	1985Ra09	PRL 54 2592 (1985)
	42 Mo 92	2760	190 ns	8+	(-)0.36	from B(E2)	not measured	1991Ha04	PR C43 2140 (1991)
	42 Mo 94	871 2956	2.8 ps 98 ns	2+ 8+	-0.13(8) or +0.01(8) 0.50(1)	A	CER TDPAD	1976Pa13 1985Ra09	PR C14 835 (1976) PRL 54 2592 (1985)
Reference isotope	42 Mo 95	0	stable	5/2+	-0.022(1)		AB	2018Py01 1982BuZE	Mol Phys 116 1328 (2018) STMP vol 96
	42 Mo 96	778	3.7 ps	2+	-0.20(8) or +0.04(8)		CER	1976Pa13	PR C14 835 (1976)
Reference isotope	42 Mo 97	0	stable	5/2+	+0.255(13)		AB	2018Py01 1982BuZE	Mol Phys 116 1328 (2018) STMP vol 96
	42 Mo 98	787	3.5 ps	2+	-0.26(9)		CER	1979Pa11	PR C20 1201 (1979)
	42 Mo 100	536	12.6 ps	2+	-0.25(7)		CER	2011Wr01	Acta Phys Pol B42 803 (2011)

Technetium *Estimation of the quadrupole coupling constant in states of the Tc atom*

Reference isotope	43 Tc 99	0	2.1 10 ⁵ y	9/2+	-0.129(6)		AB	2018Py01 1982BuZE	Mol Phys 116 1328 (2018) STMP vol 96
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Ruthenium *Calculated hyperfine structure in the 5F multiplet of the Ru atom*

	44 Ru 93	2082	2.3 μs	21/2+	+0.04(1)	[99Ru]	TDPAD	1991Ha04	PR C43 2140 (1991)
	44 Ru 96	833	2.9 ps	2+	-0.15(8)		CER	1998Hi01	PR C57 (1998)
	44 Ru 98	653	5.5 ps	2+	-0.21(8) or -0.01(9)		CER	1998Hi01	PR C57 (1998)
Reference isotope	44 Ru 99	0	stable	5/2+	+0.079(4)		AB	2018Py01	Mol Phys 116 1328 (2018)

		90	20.5 ns	3/2+	+0.231(13)	[99Ru]	ME	1982BuZE 1976Ki02	STMP vol 96 PR C13 1132 (1976)
	44 Ru 100	540	12.6 ps	2+	-0.44(4) or -0.27(7)		CER	1998Hi01	PR C57 (1998)
Reference isotope	44 Ru 101	0	stable	5/2+	+0.46(2)		AB	2018Py01 1982BuZE	Mol Phys 116 1328 (2018) STMP vol 96
	44 Ru 102	475	18 ps	2+	-0.63(4) or -0.34(3)		CER	1998Hi01	PR C57 (1998)
	44 Ru 103	0	39.2 d	3/2+	+0.62(2)	[99Ru 90 keV]	NO/S	1986Gr26	HFI 30 355 (1986)
	44 Ru 104	358	56 ps	2+	-0.78(7) or -0.20(12)		CER	1998Hi01	PR C57 (1998)
Rhodium <i>Calculation of the quadrupole coupling constants in Rh intermetallic compounds</i>									
Reference isotope	45 Rh 100	74	215 ns	(2)+	0.153 (18)		PAC	2018Py01 1996Bl15	Mol Phys 116 1328 (2018) HFI 97/98 3 (1996)
	45 Rh 103	295 357	6.6 ps 73 ps	3/2- 5/2-	-0.3(2) -0.4(2)		CERP CERP	1976Ge19 1976Ge19	ZP A 279 183 (1976) ZP A 279 183 (1976)
Palladium <i>Muonic atom X-ray hyperfine structure</i>									
	46 Pd 102	556	11.5 ps	2+	-0.20(15)		CERP	1977Fa11	NIM 146 329 (1977)
	46 Pd 104	556	9.9 ps	2+	-0.46(11)		CERP	1977Fa11	NIM 146 329 (1977)
Reference isotope	46 Pd 105	0	stable	5/2+	+0.660(11)		Mu-X	2018Py01 1978Vu01	Mol Phys 116 1328 (2018) NP A294 273 (1978)
	46 Pd 106	512	12 ps	2+	-0.51(7)		ES	1973Ho05	PRL 30 388 (1973)
	46 Pd 108	434	24 ps	2+	-0.58(4)		ES	1978Ar07	J Phys G4 961 (1978)
	46 Pd 110	374	44 ps	2+	-0.47(3)		ES	1976Li19	PR C14 952 (1976)
Silver <i>Calculation of the quadrupole coupling constant in the Ag atom.</i>									
	47 Ag 101	0	11.1 m	9/2+	+0.35(5)	[110Ag 118 keV]	CLS	1989Di12	NP A503 331 (1989)
	47 Ag 103	0	65.7 m	7/2+	+0.84(9)	[110Ag 118 keV]	CLS	1989Di12	NP A503 331 (1989)
	47 Ag 104	0	69 m	5+	+1.06(11)	[110Ag 118 keV]	CLS	1989Di12	NP A503 331 (1989)
	47 Ag 105	25	7.2 m	7/2+	+0.85(11)	[110Ag 118 keV]	CLS	1989Di12	NP A503 331 (1989)
	47 Ag 106	90	8.3 d	6+	1.11(11)	[110Ag 118 keV]	CLS	1989Di12	NP A503 331 (1989)
	47 Ag 107	93	44.3 s	7/2+	0.98(11)	[110Ag 118 keV]	LMR	1986Be01	PR C33 390 (1986)

								1984Be53	PR C30 2028 (1984)
47 Ag 108	109	438 y	6+	+1.32(7)	[110Ag 118 keV]	O		1986Be01	PR C33 390 (1986)
47 Ag 109	88	39.6 s	7/2+	(+1.02(12)	[110Ag 118 keV]	LMR		1984Be53	PR C30 2028 (1984)
	311	5.9 ps	3/2-	-0.7(3)		CER		1986Be01	PR C33 390 (1986)
	415	33 ps	5/2-	-0.3(3)		CER		1984Be53	PR C30 2028 (1984)
								1972Th16	PL 41B 585 (1972)
								1972Th16	PL 41B 585 (1972)
47 Ag 110	0	24.4 s	1+	0.24(12)		QIR		1981Do17	HFI 10 727 (1981)
Reference isomer	118	250 d	6+	+1.44(10)		O		1984Be53	PR C30 2028 (1984)
Cadmium		<i>There is no adopted reference efg for Cd.</i>							
		<i>A. Efg in 2P5/2 state of the Cd ion</i>							2013Yo02 (PRL 110 192501 (2013)
		<i>B Efg in Cd metal and dimethylcadmium.</i>							2017Ha15 Europhys Lett 117 62001 (2017)
48 Cd 101	0	1.36 m	5/2+	-0.177(7)	A	CLS		2018Yo07	PR C98 011303(R)
48 Cd 102	2718	39 ns	8+	0.76(9)	[efg Cd in Cd]	TDPAD		1992Al17	ZP A344 1 (1992)
48 Cd 103	0	7.3 m	5/2+	-0.007(3)	A	CLS		2018Yo07	PR C98 011303(R)
48 Cd 105	0	56 m	5/2+	+0.377(15)	A	OD		2018Yo07	PR C98 011303(R)
	2517	4.5 μs	21/2+	+0.88(10)	B	TDPAC		1978Sp09	HFI 4 229 (1978)
48 Cd 106	633	5.1 ps	2+	-0.28(8)			CER	1976Es02	NP A274 237 (1977)
48 Cd 107	0	6.50 h	5/2+	+0.60(2)	A	CLS		2013Yo02	PRL 110 192501 (2013)
	846	71 ns	11/2-	-0.81(10)	B	TDPAC		1978Sp09	HFI 4 229 (1978)
	2679	55 ns	21/2+	+0.91(11)	B	TDPAC		1978Sp09	HFI 4 229 (1978)
48 Cd 108	633	6.9 ps	2+	-0.45(8)			CER	1976Es02	NP A274 237 (1977)
48 Cd 109	0	461 d	5/2+	+0.60(3)	A	CLS		2013Yo02	PRL 110 192501 (2013)
	463	10.9 μs	11/2-	[-0.92(9)]	systematic extrapolation	not measured		1978Sp09	HFI 4 229 (1978)
48 Cd 110	658	5.0 ps	2+	-0.40(4)		ES		1977Gi13	J Phys G3 L169 (1977)
48 Cd 111	245	84 ns	5/2+	+0.64(3)	B	TDPAC		1978Sp09	HFI 4 229 (1978)
	396	48.5 m	11/2-	-0.75(3)	A	CLS		2017Ha15 Europhys Lett 117 62001 (2017)	
								2013Yo02	PRL 110 192501 (2013)
48 Cd 112	617	6.5 ps	2+	-0.37(4)		ES		1977Gi13	J Phys G3 L169 (1977)
48 Cd 113	264	14.1 y	11/2-	-0.61(3)	A	CLS		2013Yo02	PRL 110 192501 (2013)
48 Cd 114	558	10.2 ps	2+	-0.348(12)		ES		1981Ko06	J Phys G7 L63 (1981)
48 Cd 115	181	44.6 d	11/2-	-0.48(2)	A	CLS		2013Yo02	PRL 110 192501 (2013)

48 Cd 116	513	14.1 ps	2+	-0.42(4)		ES	1977Gi13	J Phys G3 L169 (1977)
48 Cd 117	136	3.36 h	11/2-	-0.320(13)	A	CLS	2013Yo02	PRL 110 192501 (2013)
48 Cd 119	147	2.20 m	11/2-	-0.135(6)	A	CLS	2013Yo02	PRL 110 192501 (2013)
48 Cd 121	215	8.3 s	11/2-	+0.009(6)	A	CLS	2013Yo02	PRL 110 192501 (2013)
48 Cd 123	317	1.82 s	11/2-	+0.135(7)	A	CLS	2013Yo02	PRL 110 192501 (2013)
48 Cd 125	0	0.68 s	3/2+	+0.209(10)	A	CLS	2013Yo02	PRL 110 192501 (2013)
	x	0.48 s	11/2-	+0.269(13)	A	CLS	2013Yo02	PRL 110 192501 (2013)
48 Cd 127	0	0.37 s	3/2+	+0.239(11)	A	CLS	2013Yo02	PRL 110 192501 (2013)
	x	-	11/2-	+0.34(2)	A	CLS	2013Yo02	PRL 110 192501 (2013)
48 Cd 129	0	154 ms	3/2+	+0.132(9)	A	CLS	2013Yo02	PRL 110 192501 (2013)
	x	146 ms	11/2-	+0.57(3)	A	CLS	2013Yo02	PRL 110 192501 (2013)

Indium	Calculated electric quadrupole interactions in the indium atom							2009Ya27	Can J Chem 87 802 (2009)
	49 In 104	0	1.7 m	5+	+0.63(10)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	49 In 105	0	5.07 m	9/2+	+0.79(5)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	49 In 106	0	6.2 m	7+	+0.92(6)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	49 In 107	0	32.4 min	9/2+	+0.77(5)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	49 In 108	0	58 m	7+	+0.957(7)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
		30	40 m	2+	+0.445(13)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	49 In 109	0	4.2 h	9/2+	+0.80(3)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	49 In 110	0	4.9 h	2+	+0.32(2)	[113In]	AB	1968CaZX	TH 68 Cass.
		62	69.1 m	7+	+0.95(2)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	49 In 111	0	2.83 d	9/2+	+0.76(2)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	49 In 112	0	14.9 m	1+	+0.082(5)	[113In]	AB	1968CaZX	TH 68 Cass.
		157	20.6 m	4+	+0.681(10)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
		351	0.69 μs	7+	1.00(3)	[117In 660 keV]	TDPAD	1993Io02	HFI 77 111 (1993)
		614	2.81 μs	8-	0.092(3)	[117In 660 keV]	TDPAD	1993Io02	HFI 77 111 (1993)
Reference isotope	49 In 113	0	stable	9/2+	0.761(5)		AB/MS	2018Py01 2009Ya27	Mol Phys 116 1328 (2018) Can J Chem 87 802 (2009)
	49 In 114	190	49.5 d	5+	+0.705(11)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
Reference isotope	49 In 115	0	4.4 10 ¹⁴ y	9/2+	0.772(5)		AB/MS	2018Py01 2009Ya27	Mol Phys 116 1328 (2018) Can J Chem 87 802 (2009)

	829	5.78 ns	3/2+	-0.59(4)	[117In 660 keV]	TDPAC	1973Ha61	JCP 58 3339 (1973)
49 In 116	0	14.1 s	1+	0.11(1)	[115In]	NSLR	1982Gr17	NP A386 56 (1982)
	127	54.3 m	5+	+0.764(11)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	290	2.18 s	8-	+0.296(9)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 117	0	42 m	9/2+	+0.790(10)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	660	53.6 ns	3/2+	-0.57(4)	[115In]	TDPAC	1972Ra27	PRL 28 54 (1972)
49 In 118	~60	4.45 m	5+	+0.759(8)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	~200	8.5 s	8-	+0.420(7)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 119	0	2.4 m	9/2+	+0.814(7)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	654	130 ns	3/2+	0.59(4)	[115In]	TDPAC	1980HaYW	ARHMI 1979 75 (1979)
49 In 120	0 + x	47.3 s	8-	+0.505(10)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	(70)	46.2 s	5+	+0.772(16)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 121	0	23.1 s	9/2+	+0.776(10)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 122	(40)	10.3 s	5+	+0.77(2)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	~262	10.8s	8-	+0.56(2)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 123	0	6.17 s	9/2+	+0.722(9)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 124	0	3.12 s	3+	+0.58(7)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	190	3.7 s	8-	+0.631(9)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 125	0	2.50 s	9/2+	+0.68(3)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 126	0	1.53 s	3+	+0.47(5)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
	(102)	1.64 s	8-	+0.651(11)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)
49 In 127	0	1.09 s	9/2+	+0.56(3)	[115In]	CFBLS	1987Eb02	NP A464 9 (1987)

Tin *Calculation of the quadrupole coupling constants in many molecular tin compounds.* J Phys Chem A112 1666 (2008)

A - relative to 117Sn 315 keV - calculation of quadrupole interaction in the 5p6s;3P1 state of the tin atom. At present this calculation is accurate only to, at best, +/- 10-20%.

B-relative to 116Sn 3548 keV 10+ moment estimated from theory. Accuracy estimated at 10%.

C- relative to 118Sn 3106 keV 10+ moment estimated from theory. Accuracy estimated at 10%.

50 Sn 109	0	18.0 m	5/2+	+0.33(11)	A	ABLFS	1987Eb01	ZP A326 121 (1987)
50 Sn 110	2478	5.6 ns	6+	0.30(4)	C	TDPAD	1989Vo17	IAN Ser Fiz 53 2188 (1989)
50 Sn 111	0	35.3 m	7/2+	+0.20(10)	A	ABLFS	1987Eb01	ZP A326 121 (1987)
50 Sn 112	1257	0.38 ps	2+	+0.04(9)		CER	2015Al24	PR C92 041303(R) (2015)
	2549	13.7 ns	6+	(-).0.25(5)	B	TDPAD	1975Vi03	NP A243 29 (1973)
50 Sn 113	738	86 ns	11/2-	(-).0.41(4)	B	TDPAD	1975Di02	PL B55 293 (1975)

Reference Isotope	50 Sn 114	1300 3087	0.42 ps 733 ns	2+ 7-	+0.09(8) (-)-0.32(3)	B	CER TDPAD	2015Al24 1975Di02	PR C92 041303(R) (2015) PL B55 293 (1975)
	50 Sn 115	613 714	3.26 ps 159 μs	7/2+ 11/2-	(-)-0.26(3) 0.38(6)	C	TDPAD QIR	1976Be59 1975Ri03	HFI 2 326 (1976) Phys Scr 11 228 (1975)
	50 Sn 116	1294 2366 3547	0.37 ps 348 ns 833 ns	2+ 5- 10+	+0.17(8) (-)-0.26(3) [(-)-0.41(4)]	B B	CER TDPAD not measured	2015Al24 1975Di02 1975Di02	PR C92 041303(R) (2015) PL B55 293 (1975) PL B55 293 (1975)
	50 Sn 117	315	14.0 d	11/2-	-0.42(5)	A	ABLFS	1986An24	PR C34 1052 (1986)
	50 Sn 118	1230 2321 2575 3108	0.485 ps 21.7 ns 230 ns 2.52 μs	2+ 5- 7- 10+	+0.07(9) (-)-0.22(3) 0.32(3) [0.41(4)]	B C C	CER TDPAD TDPAD not measured	2015Al24 1975Di02 1976Be59 1976Be59	PR C92 041303(R) (2015) PL B55 293 (1975) HFI 2 326 (1976) HFI 2 326 (1976)
	50 Sn 119	24 90	18.0 ns 293.1 d	3/2+ 11/2-	-0.132(1) -0.29(3)		ME ME	2018Py01 2008Ba56 1972Be79	Mol Phys 116 1328 (2018) J Phys Chem A112 1666 (2008) PL B42 349 (1972)
	50 Sn 120	1171 2284	0.64 ps 5.55 ns	2+ 5-	+0.02(7) 0.046(2)		CER TDPAC	1975Gr30 1970Wo02	PR C12 1462 (1975) ZP 232 256 (1970)
	50 Sn 121	0 6.3	27.1 h 43.9 y	3/2+ 11/2-	-0.02(2) -0.14(3)	A A	ABLFS ABLFS	1986An24 1986An24	PR C34 1052 (1986) PR C34 1052 (1986)
	50 Sn 122	1140	0.78 ps	2+	-0.08(9)		CER	2015Al24	PR C92 041303(R) (2015)
	50 Sn 123	0	129 d	11/2-	+0.03(4)	A	ABLFS	1986An24	PR C34 1052 (1986)
	50 Sn 124	1132	0.92 ps	2+	-0.08(9)		CER	2015Al24	PR C92 041303(R) (2015)
	50 Sn 125	0 28	9.64 d 9.5 m	11/2- 3/2+	+0.2(2) +0.86(8)	A A	ABLFS ABLFS	2005Le34 2004Le13	PR C72 034305 NP A734 437 (2004)
	50 Sn 126	1141	1.15 ps	2+	0.0(2)		CER	2011Al35	PR C84 1303 (2011)
	50 Sn 127	0 5	2.1 h 4.13 m	11/2- 3/2+	+0.32(14) +0.65(7)	A A	ABLFS ABLFS	2005Le34 2004Le13	PR C72 034305 NP A734 437 (2004)
	50 Sn 128	2492	2.9 μs	10+	-0.1(3)		CER	2011Al35	PR C84 1303 (2011)
	50 Sn 129	0 35	2.23 m 6.9 m	3/2+ 11/2-	+0.05(12) -0.20(19)	A A	ABLFS ABLFS	2004Le13 2005Le34	NP A734 437 (2004) PR C72 034305
	50 Sn 130	1947	1.7 m	7-	-0.39(12)	A	ABLFS	2005Le34	PR C72 034305
	50 Sn 131	0 242	56 s 58.4 s	3/2+ 11/2-	-0.04(9) 0.0(2)	A A	ABLFS ABLFS	2004Le13 2005Le34	NP A734 437 (2004) PR C72 034305

Antimony	Calculated <i>efg</i> 's in SbN, SbP, SbF and SbCl molecules							2006Ha63	J Chem Phys 125 064301 (2006)
	51 Sb 112	796	536 ns	8-	1.06(2)	[121Sb]	TDPAD	1982Ma29	PR C26 493 (1982)
	51 Sb 114	496	219 μ s	8-	1.02(16)	[121Sb]	QIR,R	1982Ma29	PR C26 493 (1982)
	51 Sb 115	2796	159 ns	19/2-	0.79(4)	[121Sb]	TDPAD	1983Se04	ZP A309 349 (1983)
	51 Sb 116	1159	10.6 ns	7+	2.5(6)	[121Sb]	TDPAD	1992Io01	ZP A343 21 (1992)
	51 Sb 117	0 3131	2.80 h 355 μ s	5/2+ (25/2)+	0.2(12) 1.14(5)	[121Sb] [121Sb]	AB QIR,R	1974Ek01 1982Ma29	NP A226 219 (1974) PR C26 493 (1982) JPhys G3 713 (1977)
		3231	290 ns	23/2-	3.7(4)	[121Sb]	TDPAD	1988Io01	PL B 200 259 (1988)
	51 Sb 118	51 270 965	20.6 μ s 13.4 ns 22.6 ns	(3)+ 3- 7+	0.9(2) 0.39(8) 2.6(5)	[121Sb] [121Sb] [121Sb]	TDPAD TDPAD TDPAD	1982Ma29 1985Di07 1988Io01	PR C26 493 (1982) ZP A320 613 (1985) PL B 200 259 (1988)
	51 Sb 119	2554	130 ns	19/2-	3.18(13)	[121Sb]	TDPAD	1991Io02	NP A531 112 (1991)
	51 Sb 120	78	246 ns	3+	0.63(2)	[121Sb]	TDPAD	1982Ma29	PR C26 493 (1982)
Reference Isotope	51 Sb 121	0	stable	5/2+	-0.543(11)		O	2006Ha63	J Chem Phys 125 064301 (2006)
		37	3.5 ns	7/2+	-0.727(16)	[121Sb]	ME	1978Bu24 1970St13	ZP A288 247 (1978) PL A32 91 (1970)
	51 Sb 122	0 61	2.72 d 1.86 μ s	2- 3+	+1.28(8) 0.63(2)	[121Sb] [121Sb]	O TDPAD	1960Fe08 1982Ma29	PhMg 5 1309 (1960) PR C26 493 (1982)
Reference Isotope	51 Sb 123	0	stable	7/2+	-0.692(14)		O	2006Ha63 1978Bu24	J Chem Phys 125 064301 (2006) ZP A288 247 (1978)
	51 Sb 124	0	60.2 d	3-	+2.8(2)	[121Sb]	NO/S	1985He16	ZP A322 281 (1985)
Tellurium	There is no adopted reference <i>efg</i> for Te.								
	A. <i>Efg</i> in the lased state of the Te atom calculated by semi-empirical methods								
	52 Te 120	560 1162	9.3 ps 3.2 ps	2+ 4+	-0.41(3) -0.77(19)		CER CER	2018Sa37 2018Sa37	Acta Phys Pol B49 541 (2018) Acta Phys Pol B49 541 (2018)
	52 Te 122	564	7.46 ps	2+	-0.57(5)		CER	1976Bo12	NP A261 498 A261
	52 Te 124	603	6.2 ps	2+	-0.45(5)		CER	1976Bo12	NP A261 498 A261
	52 Te 125	36 145 321	1.48 ns 57.4 d 673 ps	3/2+ 11/2- 9/2-	-0.31(2) 0.0(2) 0.12(7)	[129] A [125Te 36 keV]	ME CLS IPAC	1977La03 2006Si40 1976Va28	PR B15 2504 HFI 171 173 (2006) HFI 2 321 (1976)
	52 Te 126	666	4.52 ps	2+	-0.23(5)		CER	1976Bo12	NP A261 498 A261

	52 Te 127	88	106 d	11/2-	0.17(12)	A	CLS	2006Si40	HFI 171 173 (2006)
	52 Te 128	743	3.3 ps	2+	-0.22(5)		CER	1976Bo12	NP A261 498 A261
	52 Te 129	0 106	69.6 m 33.6 d	3/2+ 11/2-	0.055(13) 0.40(3)	[129I] A	NO/ME CLS	1987Be36 2006Si40	HFI 35 1023 (1987) HFI 171 173 (2006)
	52 Te 130	840	2.3 ps	2+	-0.12(5)		CER	1976Bo12	NP A261 498 A261
	52 Te 131	182	33.3 h	11/2-	0.25(14)	A	CLS	2006Si40	HFI 171 173 (2006)
	52 Te 133	0 334	12.5 m 55.4 m	3/2+ 11/2-	0.23(9) 0.28(14)	A A	CLS CLS	2006Si40 2006Si40	HFI 171 173 (2006) HFI 171 173 (2006)
	52 Te 135	0	19.0 s	7/2-	0.29(9)	A	CLS	2006Si40	HFI 171 173 (2006)
Iodine	<i>Calculated efg of states in the I atom.</i>							2013Ch52	J Phys Chem A117 12616 (2013)
<i>Reference isotope</i>	53 I 125	0	59.4 d	5/2+	-0.752(17)	[127I]	MA	1958FI39	PR 110 536 (1958)
	53 I 127	0	stable	5/2+	-0.688(10)		AB	1976Fu06	JPCR 5 835 (1976)
		58	1.95 ns	7/2+	-0.617(11)	[127I]	ME	2013Ch52 1964Pe15	J Phys Chem A117 12616 (2013) PL 13 198 (1964)
	53 I 129	0 28	1.57 10 ⁷ y 16.8 ns	7/2+ 5/2+	-0.483(8) -0.598(10)	[127I] [127I]	Q,MA ME	1953Li16 1972Ro41	PR 90 609 (1953) NIM 105 509 (1972)
	53 I 131	0 150 1797	8.03 d 0.95 ns 5.9 ns	7/2+ 5/2+ (15/2)-	-0.34(2) 0.30(1) 0.66(6)	[127I] [129I] [129I 28 keV]	AB TDPAC TDPAC	1960Li13 2020Al17 1973Ha61	PR 119 2022 (1960) Eur Phys J A56 269 (2020) JCP 58 3339 (1973)
	53 I 132	0 50 278	2.30 h 7.14 ns 1.42 ns	4+ 3+ 1+	0.08(1) 0.25(2) -0.148(5)	[127I] [129I] [129I]	AB TDPAC TDPAC	1960Wh06 2020Al17 1979Oo01	BAPS 5 504 (1960) Eur Phys J A56 269 (2020) NP A321 180 (1979)
	53 I 133	0	20.8 h	7/2+	-0.23(1)	[127I]	AB	1961Al20	UCRL 9850 (1960)
Xenon	<i>Calculated efg in Xe molecular systems</i>							2016Ca 44	Chem Phys Lett 660 228 (2016)
	<i>A - Q estimated from B(E2)</i>								
	<i>B - Efg estimated from systematics in Te metal</i>								
	54 Xe 117	0	1.02 m	5/2+	+1.14((4)	[131Xe]	CLS	1990NeZY	PC Neugart (1990)
	54 Xe 119	0	5.8 m	5/2+	+1.29(5)	[131Xe]	CLS	1990NeZY	PC Neugart (1990)
	54 Xe 121	0	40.1 m	5/2+	+1.31(5)	[131Xe]	CLS	1990NeZY	PC Neugart (1990)
	54 Xe 123	185 206	5.5 μs 11.8 ns	7/2(-) 9/2-	1.4(3) 1.1(6)	[125Xe 296 keV] [125Xe 296 keV]	TDPAD TDPAD(AMPL)	1982Ze05 1982Ze05	ZP A308 227 (1982) ZP A308 227 (1982)

Reference isotope	54 Xe 125	253 296	57 s 140 ns	9/2- 7/2+	+0.417(15) 1.40(15)	[131Xe] A	CLS not measured	1990NeZY 1982Ze05	PC Neugart (1990) ZP A308 227 (1982)
	54 Xe 127	297	1.15 m	9/2-	+0.68(2)	[131Xe]	CLS	1990NeZY	PC Neugart (1990)
	54 Xe 129	40 236	0.97 ns 8.88 d	3/2+ 11/2-	-0.393(10) +0.63(2)	[131Xe] [131Xe]	ME CLS	1964Pe06 1990NeZY	PR 135B 1102 (1964) PC Neugart (1990)
	54 Xe 130	536 1205	8.6 ps 2.4 ps	2+ 4+	-0.38(16) -0.41(12)		CER CER	2020Mo35 2020Mo35	PR C102 054304 (2020) PR C102 054304 (2020)
	54 Xe 131	0 164	stable 11.8 d	3/2+ 11/2-	-0.114(1) +0.72(3)	[131Xe]	CLS CLS	1989Bo03 1990NeZY	PL B216 7 (1989) PC Neugart (1990)
	54 Xe 132	2214	87 ns	7-	0.010(5)	B	TDPAD	1987Le31	UkrF 32 1636 (1987)
	54 Xe 133	0 233	5.25 d 2.20 d	3/2+ 11/2-	+0.140(5) +0.76(5)	[131Xe] [131Xe]	CLS CLS	1990NeZY 1990NeZY	PC Neugart (1990) PC Neugart (1990)
	54 Xe 135	0 527	9.14 h 15.3 m	3/2+ 11/2-	+0.210(7) +0.61(2)	[131Xe] [131Xe]	CLS CLS	1990NeZY 1990NeZY	PC Neugart (1990) PC Neugart (1990)
	54 Xe 137	0	3.82 m	7/2-	-0.47(2)	[131Xe]	CLS	1989Bo03	PL B216 7 (1989)
	54 Xe 139	0	39.7 s	3/2-	+0.39(2)	[131Xe]	CLS	1989Bo03	PL B216 7 (1989)
	54 Xe 141	0	1.73 s	5/2+	-0.57(2)	[131Xe]	CLS	1989Bo03	PL B216 7 (1989)
	54 Xe 143	0	0.51 s	5/2-	+0.91(3)	[131Xe]	CLS	1989Bo03	PL B216 7 (1989)

Caesium	Calculated efg in CsF molecule.							2008Py02	Mol Phys 106 1965 (2008)
	A - estimated efg at Cs in Ga metal								
	55 Cs 118	0	14 s	2	+1.31(17)	[133Cs]	ABLS	1987Co19	NP A468 1 (1987)
	55 Cs 119	0 0 + x	43 s 30.4 s	9/2+ 3/2+	+2.65(17) +0.85(12)	[133Cs] [133Cs]	ABLS ABLS	1981Th06 1981Th06	NP A367 1 (1981) NP A367 1 (1981)
	55 Cs 120	0	61.3 s	2+	+1.36(7)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 121	0 68	155 s 122 s	3/2+ 9/2+	+0.79(4) +2.53(13)	[133Cs] [133Cs]	ABLS ABLS	1981Th06 1981Th06	NP A367 1 (1981) NP A367 1 (1981)
	55 Cs 122	0 ~ 140	21 s 3.7 m	1+ 8-	-0.179(10) +3.09(8)	[133Cs] [133Cs]	ABLS ABLS	1981Th06 1981Th06	NP A367 1 (1981) NP A367 1 (1981)
	55 Cs 124	0	31 s	1+	-0.69(4)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 126	0	1.64 m	1+	-0.64(3)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)

Reference isotope	55 Cs 127	66	24.9 ns	5/2(+)	0.58(12)	A	TDPAC	1999Co22	NIM B152 357 (1999)
	55 Cs 128	0	3.6 m	1+	-0.54(3)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 130	0	29.2 m	1+	-0.056(6)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
		0+x	3.46 m	5(-)	+1.36(8)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 131	0	9.69 d	5/2+	+0.59(2)	[133Cs]	ABLS	1975Ac01	NP A248 157 (1975)
		134	8.6 ns	5/2+	0.20(2)	[133Cs 81 keV]	TDPAC	2000De13	Eur Phys J A7 177 (2000)
	55 Cs 132	0	6.48 d	2(-)	+0.48(2)	[133Cs]	ABLS	1975Ac01	NP A248 157 (1975)
	55 Cs 133	0	stable	7/2+	-0.00343(10)		MB	1998Pe18	JCP 108 6739 (1998)
		81	6.28 ns	5/2+	0.30(2)	[133Cs]	ME	1977Ca30	PR B15 3318 (1977)
	55 Cs 134	0	2.06 y	4+	+0.37(2)	[133Cs]	ABLS	1975Ac01	NP A248 157 (1975)
		139	2.91 h	8-	+0.92(8)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 135	0	2.3 10 ⁶ y	7/2+	+0.048(3)	[133Cs]	ABLS	1975Ac01	NP A248 157 (1975)
		1633	53 m	19/2-	+0.83(7)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 136	0	13.2 d	5+	+0.213(15)	[133Cs]	ABLS	1975Ac01	NP A248 157 (1975)
		0+x	19 s	8-	+0.70(3)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 137	0	30.1 y	7/2+	+0.048(2)	[133Cs]	ABLS	1975Ac01	NP A248 157 (1975)
	55 Cs 138	0	33.4 m	3-	+0.112(17)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
		80	2.9 m	6-	-0.37(5)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 139	0	9.3 m	7/2+	-0.063(14)	[133Cs]	ABLS	1979Bo01	ZP A289 227 (1979)
	55 Cs 140	0	63.7 s	1-	-0.094(15)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 141	0	24.8 s	7/2+	-0.42(7)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 143	0	1.79 s	3/2+	+0.44(3)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 144	0	0.99 s	1(-)	+0.29(2)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 145	0	0.59 s	3/2+	+0.58(6)	[133Cs]	ABLS	1981Th06	NP A367 1 (1981)
	55 Cs 146	0	0.32 s	1-	+0.21(3)	[133Cs]	ABLS	1987Co19	NP A468 1 (1987)

Barium	<i>Efg calculations in the d states of Ba+</i>							2018Py01	Mol Phys 116 1328 (2018)
	56 Ba 121	0	30 s	5/2(+)	+1.89(13)	[135Ba]	CLS	1988We14	PL B211 272 (1988)
	56 Ba 123	0	2.7 m	5/2+	+1.57(13)	[135Ba]	CLS	1988We14	PL B211 272 (1988)
	56 Ba 127	80	1.9 s	7/2(-)	+1.71(14)	[135Ba]	CLS	1992Da06	J Phys G 17 L67 (1992)

	56 Ba 129	8.4	2.14 h	7/2+	+1.68(14)	[135Ba]	CLS	1979Be25	ZP A291 219 (1979)
	56 Ba 130	357 2476	42 ps 9.4 ms	2+ 8-	-1.02(15) or -0.09(15) +2.31(6)	[135Ba]	CER CLS	1989Bu07 2002Mo31	NP A494 102 (1989) PL B547 200 (2002)
	56 Ba 131	188	14.6 m	9/2-	+1.54(14)	[135Ba]	CLS	1983Mu12	NP A403 234 (1983)
	56 Ba 133	288	38.9 h	11/2-	+0.92(6)	[135Ba]	CLS	1979Be25	ZP A291 219 (1979)
	56 Ba 134	605	5.1 ps	2+	-0.26(12) or +0.15(12)		CER	1989Bu07	NP A494 102 (1989)
Reference isotope	56 Ba 135	0	stable	3/2+	+0.153(2)		CFBLS	1984We15	ZP A318 125 (1984)
		268	28.7 h	11/2-	+0.99(15)	[135Ba]	CLS	1979Be25	ZP A291 219 (1979)
	56 Ba 136	819	1.93 ps	2+	-0.19(6) or +0.07(7)		CER	1986Ro15	PR C34 732 (1986)
Reference isotope	56 Ba 137	0	stable	3/2+	+0.236(3)		CFBLS	1984We15	ZP A318 125 (1984)
		662	2.55 m	11/2-	+0.82(10)	[135Ba]	CLS	1983Mu12	NP A403 234 (1983)
	56 Ba 138	1436	0.20 ps	2+	-0.14(6) or +0.08(6)		CER	1989Bu07	NP A494 102 (1989)
	56 Ba 139	0	83.1 m	7/2-	-0.552(13)	[135Ba]	CFBLS	1988We07	ZP A329 407 (1988)
	56 Ba 140	602	9 ps	2+	-0.5(3)		CER	2012Ba40	PR C86 034310 (2012)
	56 Ba 141	0	18.3 m	3/2-	+0.437(10)	[135Ba]	CFBLS	1988We07	ZP A329 407 (1988)
	56 Ba 143	0	14.5 s	5/2(+)	-0.85(2)	[135Ba]	CFBLS	1988We07	ZP A329 407 (1988)
	56 Ba 145	0	4.31 s	5/2(-)	+1.18(2)	[135Ba]	CFBLS	1988We07	ZP A329 407 (1988)
Lanthanum								2011It09	Theor Chem Acc 129 409 (2011)
Calculated <i>efg</i> in 1D2 state of La+.								2020La06	PR C101 034315 (2020)
A Calculated <i>efg</i> for La in Te metal.									
	57 La 133	536	62 ns	11/2-	1.7(3)	A	TDPAD	2020La06	PR C101 034315 (2020)
	57 La 135	0	19.5 h	5/2+	-0.4(4)	[139La]	CLS	2003Ii03	PR C68 054328 (2003)
	57 La 137	0	6 10 ⁴ y	7/2+	+0.21(4)	[139La]	CLS	2003Ii03	PR C68 054328 (2003)
	57 La 138	0	1.0 10 ¹¹ y	5+	+0.39(3)	[139La]	CLS	2003Ii03	PR C68 054328 (2003)
Reference isotope	57 La 139	0	stable	7/2+	+0.206(4)		LRS	2006Sc30 2011It09	Physica Scripta 73 217 (2006) Theor Chem Acta 129 409 (2011)
	57 La 140	0	40.3 h	3-	+0.087(13)	[139La]	NO/S	1966BI05	PR 143 911 (1966)
Cerium									
There is no adopted <i>efg</i> calculation for cerium.									
A. Normalised to nuclear model estimate of Q 138Ce 3538 keV.									
	58 Ce 129	108	60 ns	(7/2-)	1.32(13)	A	TDPAD	1998Io01	NP A633 459 (1998)

58 Ce 130	2454	100 ns	7-	1.8(2)	A	TDPAD	1999Io02	PR C60 024316 (1999)
58 Ce 131	162	88 ns	9/2-	0.92(10)	A	TDPAD	1998Io01	NP A633 459 (1998)
58 Ce 134	3209	308 ns	10+	+1.32(12)	A	TDPAD	1983Da29	HFI 15/16 101 (1983)
58 Ce 136	3095	2.2 μ s	10+	+1.11(11)	A	TDPAD	1983Da29	HFI 15/16 101 (1983)
58 Ce 138	3539	81 ns	10+	estimated +0.77 eb		not measured	1983Da29	HFI 15/16 101 (1983)
58 Ce 140	2083	3.4 ns	4+	0.35(7)	[139La]	TDPAC	1973KIZV	JPSJ 34 265 (1973)
58 Ce 142	641	5.6 ps	2+	-0.16(5) or -0.37(5)		CER	1988Ve08 1989Sp07	PR C38 2982 (1988) AuJP 42 345 (1989)

Praseodymium *Efg calculated in the Pr ion with estimated Sternheimer correction (1994II01)*

N.B. Deviation from standard adopted by Pyykko (2008Py02) who gives Q ¹⁴¹Pr -0.059(4) b.

<i>Reference isotope</i>	59 Pr 141	0	stable	5/2+	-0.077(6)		CLS	1994Ii01	PR C50 661 (1994)
	59 Pr 142	0	19.1 h	2-	0.039(17)	[141Pr]	AB	1962Ca10	PR 126 1004 (1962)
	59 Pr 143	0	13.57 d	7/2+	+0.77(16)	[141Pr]	CLS	1994Ii01	PR C50 661 (1994)

Neodymium *Efg calculated in the Nd ion with estimated Sternheimer correction*

N.B. Deviation from standard adopted by Pyykko (2008Py02) who gives Q ¹⁴³Nd -0.63(6) b.

	60 Nd 135	0	12.4 m	9/2-	+1.9(5)	[143Nd]	CLS	1992Le09	JPhys G18 1177 (1992)
	60 Nd 139	0	29.7 m	3/2+	+0.28(9)	[143Nd]	CLS	1992Le09	JPhys G18 1177 (1992)
	60 Nd 141	0	2.49 h	3/2+	+0.32(13)	[143Nd]	CLS	1992Le09	JPhys G18 1177 (1992)
<i>Reference isotope</i>	60 Nd 143	0	stable	7/2-	-0.61(2)		AB	1992Au04	ZP D23 19 (1992)
	60 Nd 144	697	2.97 ps	2+	-0.15(6) or -0.28(6)		CER	1989Sp07	AuJP 42 345 (1989)
	60 Nd 145	0	stable	7/2-	-0.314(12)	[143Nd]	AB	1992Au04	ZP D23 19 (1992)
	60 Nd 146	454	20.9 ps	2+	-0.78(9)		CER	1970Ge08	NP A151 252 (1970)
	60 Nd 147	0	11.0 d	5/2-	+0.9(3)	[143Nd]	AB	1970PiZR	BAPS 15 769 (1970)
	60 Nd 148	302	80 ps	2+	-1.46(13)		CER	1970Ge08	NP A151 252 (1970)
	60 Nd 149	0	1.73 h	5/2-	+1.3(3)	[143Nd]	AB	1970PiZR	BAPS 15 769 (1970)
	60 Nd 150	130	1.48 ns	2+	-2.0(5)		CER	1970Ge08	NP A151 252 (1970)

Promethium *Empirical efg estimate in Pm atom*

Reference isotope	61 Pm 145	0	17.7 y	5/2+	+0.23(8)	[147Pm]	CLS	1992Al03	JP B25 571 (1992)
	61 Pm 147	0	2.623 y	7/2+	+0.74(20)		O	1966Re04	PR 141 1123 (1966)
	61 Pm 148	0	5.37 d	1-	+0.2(2)	[147Pm]	AB	1963Bu14	PR 132 723 (1963)
	61 Pm 151	0	28.4 h	5/2 +	2.2(9)	[147Pm]	AB	1963Bu14	PR 132 723 (1963)
Samarium <i>Muonic atom X-ray hyperfine structure</i>									
Reference isotope	62 Sm 140	3172	19.4 ns	10+	1.7(5)	[154Sm 82]	TDPAD	1985Be23	ZP A321 403 (1985)
	62 Sm 141	176	22.6 m	11/2-	+1.6(5)	[147Sm]	CLS	1992Le09	JPhys G18 1177 (1992)
	62 Sm 142	2372	170 ns	7-	+1.1(3)	[154Sm 82]	TDPAD	1985Be23 1986Da22	ZP A321 403 (1985) PL B181 21 (1986)
	62 Sm 143	0	8.75 m	3/2+	+0.4(2)	[147Sm]	CLS	1992Le09	JPhys G18 1177 (1992)
	62 Sm 145	0	340 d	7/2-	-0.60(7)	[147Sm]	LRFS	1990En01	JPhys G16 105 (1990)
	62 Sm 147	0	1.1 10 ¹¹ y	7/2-	-0.26(3)		Mu-X	2008Py02 1981Ba28	Mol Phys 106 1965 (2008) NP A364 446 (1981)
		121	0.80 ns	5/2-	-0.5(2)	[147Sm]	ME	1971Pa04	PR C3 841 (1971)
	62 Sm 148	550	7.7 ps	2+	-1.0(3)		CER	1989Ra17	JPSJ 34 443 (1973)
	62 Sm 149	0	stable	7/2-	+0.075(8)	[147Sm]	AB	1972Ch55 1992Le09	PR A6 2011 (1972) JPhys G18 1177 (1992)
		23	7.3 ns	5/2-	+1.01(9)	[147Sm]	Mu-X	1981Ba28	NP A364 446 (1981)
	62 Sm 150	334	48 ps	2+	-1.3(2)		CER	1973Gr06	PRL 30 453 (1973)
	62 Sm 151	0	90 y	5/2-	+0.71(7)	[147Sm]	LRFS	1990En01	JPhys G16 105 (1990)
	62 Sm 152	122	1.40 ns	2+	-1.666(16)		Mu-X	1979Po05	NP A316 295 (1979)
	62 Sm 153	0	46.3 h	3/2+	+1.30(12)	[147Sm]	LRFS	1990En01	JPhys G16 105 (1990)
Reference isotope	62 Sm 154	82	3.0 ns	2+	-1.87(4)		Mu-X	1979Po05	NP A316 295 (1979)
	62 Sm 155	0	22.3 m	3/2-	1.13(13)	[147Sm]	AB	1976Fu06	JPCR 5 835 (1976)
Europium <i>Muonic atom X-ray hyperfine structure</i>									
	63 Eu 140	0 + x	1.51 s	1(+)	+0.31(4)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 141	0	40.7 s	5/2+	+0.85(4)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 142	0	2.4 s	1+	+0.12(5)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)

		0 + x	1.22 m	8-	+1.41(6)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 143	0	2.6 m	5/2+	+0.51(3)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 144	0	10 s	1+	+0.10(3)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 145	0	5.93 d	5/2+	+0.29(2)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 146	0	4.61 d	4-	-0.18(6)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 147	0	24.1 d	5/2+	+0.55(3)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 148	0	54.5 d	5-	+0.35(6)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 149	0	93.1 d	5/2+	+0.75(2)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 150	0	36.9 y	5(-)	+1.13(5)	[153Eu]	CLS	1985Ah02	ZP A321 35 (1985)
	63 Eu 151	0	stable	5/2+	+0.903(10)	[153Eu]	Mu-X	1984Ta04	PR C29 1830 (1984)
		22	9.6 ns	7/2+	+1.28(2)		Mu-X	1984Ta05	PR C29 1897 (1984)
	63 Eu 152	0	13.52 y	3-	+2.72(3)	[153Eu]	CLS	1986Al33	YadF 44 1134 (1986)
Reference isotope	63 Eu 153	0	stable	5/2+	+2.41(2)		Mu-X	1984Ta04	PR C29 1830 (1984)
		83	0.79 ns	7/2+	+0.44(2)		Mu-X	1984Ta05	PR C29 1897 (1984)
		103	3.85 ns	3/2+	+1.253(12)	[153Eu]	ME	1973Ar19	PL A44 279 (1973)
	63 Eu 154	0	8.6 y	3-	+2.85(10)	[153Eu]	CLS	1986Al33	YadF 44 1134 (1986)
	63 Eu 155	0	4.75 y	5/2+	+2.5(3)	[153Eu]	CLS	1990Al34	ZP A337 257 (1990)
	63 Eu 157	0	15.2 h	5/2+	+2.6(3)	[153Eu]	CLS	1990Al34	ZP A337 257 (1990)
	63 Eu 158	0	45.9 m	1(-)	+0.66(14)	[153Eu]	CLS	1990Al34	ZP A337 257 (1990)
	63 Eu 159	0	18.1 m	5/2+	+2.7(3)	[153Eu]	CLS	1990Al34	ZP A337 257 (1990)

Gadolinium

Muonic atom X-ray hyperfine structure

64 Gd 144	3433	145 ns	10+	-1.40(6)	[155Gd]	TDPAD	1982Ha20 1985Da20	NP A379 287 (1982) NP A443 135 (1985)
64 Gd 147	997	21.4 ns	13/2+	-0.70(8)	[155Gd]	TDPAD	1982Ha20 1985Da20	NP A379 287 (1982) NP A443 135 (1985)
	3582	27 ns	27/2-	-1.21(9)	[155Gd]	TDPAD	1982Ha20 1985Da20	NP A379 287 (1982) NP A443 135 (1985)
	8587	510 ns	49/2+	-3.00(18)	[155Gd]	TDPAD	1982Ha20 1985Da20	NP A379 287 (1982) NP A443 135 (1985)
64 Gd 148	2695	16.6 ns	9-	0.96(5)	[155Gd]	TDPAD	1982Ha20	NP A379 287 (1982)

	64 Gd 154	123	1.18 ns	2+	-1.82(4)		Mu-X	1983La08	PR C27 1772 (1983)
Reference isotope	64 Gd 155	0	stable	3/2-	+1.27(3)		Mu-X	1983La08	PR C27 1772 (1983)
		87	6.50 ns	5/2+	+0.110(8)	[155Gd]	ME	1974Ar23	NP A233 385 (1974)
		105	1.16 ns	3/2+	+1.27(5)	[155Gd]	ME	1974Ar23	NP A233 385 (1974)
	64 Gd 156	89	2.21 ns	2+	-1.93(4)		Mu-X	1983La08	PR C27 1772 (1983)
Reference isotope	64 Gd 157	0	stable	3/2-	+1.35(3)		Mu-X	1983La08	PR C27 1772 (1983)
		64	0.46 μ s	5/2+	+2.43(7)	[157Gd]	ME	1974Ar23	NP A233 385 (74)
	64 Gd 158	80	2.52 ns	2+	-2.01(4)		Mu-X	1983La08	PR C27 1772 (1983)
	64 Gd 160	75	2.72 ns	2+	-2.08(4)		Mu-X	1983La08	PR C27 1772 (1983)
Terbium									
<i>Muonic atom X-ray hyperfine structure</i>									
<i>A. Efg estimate at Tb in yttrium ethylsulfate</i>									
	65 Tb 148	0	60 m	2-	-0.3(2)	[159Tb]	CLS	1990Al36	ZP A337 367 (1990)
	65 Tb 150	0 + x	3.48 h	2(-)	0.00(13)	[159Tb]	CLS	1990Al36	ZP A337 367 (1990)
	65 Tb 152	0	17.5 h	2-	+0.34(13)	[159Tb]	CLS	1990Al36	ZP A337 367 (1990)
	65 Tb 153	0	2.34 d	5/2+	+1.08(14)	[159Tb]	CLS	1990Al36	ZP A337 367 (1990)
	65 Tb 154	0 + x	9.4 h	3-	+2.4(13)	[159Tb]	NO/S	1983Be03	JPhys G9 213 (1983)
	65 Tb 155	0	5.32 d	3/2+	+1.41(6)	[159Tb]	CLS	1990Al36	ZP A337 367 (1990)
	65 Tb 156	0	5.35 d	3-	+2.3(8)	[159Tb]	NO/S	1979Ri17 1983Be03	CzJP B29 620 (1979) JPhys G9 213 (1983)
	65 Tb 157	0	71 y	3/2+	+1.40(8)	[159Tb]	CLS	1990Al36	ZP A337 367 (1990)
	65 Tb 158	0	180 y	3-	+2.7(5)	A	EPR/NO/S	1968Ea04	PR 170 1083 (1968)
Reference isotope	65 Tb 159	0	stable	3/2+	+1.432(8)		Mu-X	1984Ta04	PR C29 1830 (1984)
	65 Tb 160	0	72.3 d	3-	+3.85(5)	[159Tb]	NMR/ON	1987Ma42	PRL 59 1764 (1987)
	65 Tb 161	0	6.89 d	3/2+	+1.3(6)	[159Tb]	NO/S	1983Ri15	HFI 15 83 (1983)
Dysprosium									
<i>Muonic atom X-ray hyperfine structure</i>									
<i>A. Unpublished results from Neugart, PC to Raghavan (1987). No information on interaction analysis.</i>									
<i>B. Analysis of perturbation of TDPAC in liquid sources</i>									
	66 Dy 147	751	55 s	(11/2-)	+0.67(10)	A	CLS	1989Ra17	ADNDT 42 189 (1989)
	66 Dy 149	0	4.2 m	7/2-	-0.62(5)	A	CLS	1989Ra17	ADNDT 42 189 (1989)

	66 Dy 151	0	17.9 m	7/2-	-0.30(5)	A	CLS	1989Ra17	ADNDT 42 189 (1989)
	66 Dy 153	0	6.4 h	7/2-	-0.15(9)	[163Dy]	AB	1973Ek01	PS 7 31 (1973)
	66 Dy 155	0	9.9 h	3/2-	+0.96(2)	[163Dy]	AB	1973Ek01	PS 7 31 (1973)
	66 Dy 157	0	8.14 h	3/2-	+1.29(2)	[163Dy]	AB	1973Ek01	PS 7 31 (1973)
	66 Dy 159	0	144 d	3/2-	+1.37(2)	A	CLS	1989Ra17	ADNDT 42 189 (1989)
	66 Dy 160	87	2.02 ns	2+	1.8(4)	B	TDPAC	1970Wa25	ZP A238 35 (1970)
	66 Dy 161	0	stable	5/2+	+2.51(2)	[163Dy]	AB	1974Fe05	PL A49 287 (1974)
		26	29.1 ns	5/2-	+2.51(2)	[161Dy]	ME	1973St23	JPCR 5 1093 (1973)
		44	0.83 ns	7/2+	+0.53(13)	[161Dy]	ME	1973Sy01	PR C7 2056 (1973)
		75	3.14 ns	3/2-	+1.45(6)	[161Dy]	ME	1973St23	JPCR 5 1093 (1973)
Reference isotope	66 Dy 163	0	stable	5/2-	+2.65(2)		Mu-X	1984Ta04	PR C29 1830 (1984)
	66 Dy 164	73	2.39 ns	2+	-2.08(15)	[161Dy]	ME	1968Mu01	ZP A208 184 (1968)
	66 Dy 165	0	2.33 h	7/2+	+3.48(7)	[161Dy]	AB	1968 Ra03	PR 165 1360 (1968)

Holmium	Pionic atom X-ray hyperfine structure				n				
	67 Ho 152	0	161.8 s	2-	+0.1(2)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
		160	49.8 s	9+	-1.3(8)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 153	0	2.01 m	11/2-	-1.1(5)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 154	0	11.76 m	2-	+0.19(10)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
		320	3.10 m	8+	-1.0(5)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 155	0	48 m	5/2+	+1.56(12)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 156	0	56 m	4(+)	+2.40(18)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 157	0	12.6 m	7/2-	+3.05(13)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 158	0	11.3 m	5+	+4.2(4)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
		67.2	28 m	2-	+1.66(17)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 159	0	33.05 m	7/2-	+3.27(13)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 160	0	25.6 m	5+	+4.0(2)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
		60	5.02 h	2-	+1.83(17)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 161	0	2.48 h	7/2-	+3.30(11)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)
	67 Ho 162	106	67 m	6-	+4.0(7)	[165Ho]	LRIS	1989AI27	NP A504 549 (1989)

	67 Ho 163	0	4570 y	7/2-	+3.7(6)	[165Ho]	LRIS	1989Al27	NP A504 549 (1989)
Reference isotope	67 Ho 165	0	stable	7/2-	+3.58(2)		Pi-X	1983OI03	NP A403 572 (1983)
		95	22 ps	9/2-	+3.52(4)	[165Ho]	Mu-X	1976Po05	NP A262 493 (1976)
Erbium	Muonic atom X-ray hyperfine structure								
	A - Estimated efg in Er metal.								
	68 Er 153	0	37.1 s	(7/2-)	-0.42(2)	[167Er]	CLS	1987OtZW	CERN EP 87/51 (1987)
	68 Er 155	0	5.3 m	7/2-	-0.27(2)	[167Er]	CLS	1987OtZW	CERN EP 87/51 (1987)
	68 Er 157	0	18.7 m	3/2-	+0.92(1)	[167Er]	CLS	1987OtZW	CERN EP 87/51 (1987)
	68 Er 159	0	36 m	3/2-	+1.17(1)	[167Er]	CLS	1987OtZW	CERN EP 87/51 (1987)
	68 Er 161	0	3.21 h	3/2-	+1.363(8)	[167Er]	AB	1972Ek03	NP A194 237 (1972)
	68 Er 162	102	1.29 ns	2+	<0		CER	1981Hu02	PR C23 240 (1981)
		901	1.25 ps	2+	1.8(6)		CER	1983Hu01	PR C27 550 (1983)
	68 Er 163	0	75.0 m	5/2-	+2.56(2)	[167Er]	CLS	1987OtZW	CERN EP 87/51 (1987)
	68 Er 164	92	1.48 ns	2+	<0		CER	1981Hu02	PR C23 240 (1981)
		860	1.9 ps	2+	2.4(3)		CER	1983Hu01	PR C27 550 (1983)
	68 Er 165	0	10.36 h	5/2-	+2.71(3)	[167Er]	CLS	1987OtZW	CERN EP 87/51 (1987)
	68 Er 166	81	1.81 ns	2+	-1.9(4)	A	ME	1965Hu01	ZP 182 499 (1965)
		265	118 ps	4+	-2.7(9)		CER	1970McZQ	ORNL 4513 56 (1970)
		786	3.1 ps	2+	2.2(3)		CER	1983Hu01	PR C27 550 (1983)
Reference isotope	68 Er 167	0	stable	7/2+	+3.57(3)		Mu-X	1984Ta04	PR C29 1830 (1984)
	68 Er 168	264	114 ps	4+	-2.2(10)		CER	1970McZQ	ORNL 4513 56 (1970)
		821	2.8 ps	2+	2.3(2)		CER	1983Hu01	PR C27 550 (1983)
	68 Er 170	79	1.89 ns	2+	-1.9(2)		CER	1973Lu02	PR C8 391 (1973)
		260	-	4+	-2.2(10)		CER	1970McZQ	ORNL 4513 56 (1970)
934		1.81 ps	2+	2.0(3)		CER	1983Hu01	PR C27 550 (1983)	
	68 Er 171	0	7.52 h	5/2-	2.86(9)	[167Er]	AB	1964Bu09	PR 135 B1281 (1964)
Thulium	There is no adopted reference efg for Tm.								
	A. For details of the efg used see 1973Ek01/1988Al04								
	B. Includes estimated Sternheimer correction								
	69 Tm 153	0	1.48 s	(11/2-)	+0.5(10)	[169Tm]	LRIS	2000Ba16	PR C61 034304 (2000)
	69 Tm 154	0	8.1 s	(2-)	+0.4(9)	A	LRIS	2000Ba16	PR C61 034304 (2000)
		0 + x	3.30 s	(9+)	-0.2(4)	A	LRIS	2000Ba16	PR C61 034304 (2000)

69 Tm 156	0	1.40 m	2-	-0.48(11)	A	LRIS	1987AlZb	LINPR 1309 (1987)
69 Tm 158	0	3.98 m	2-	+0.74(11)	A	LRIS	1988AlO4	NP A477 37 (1988)
69 Tm 159	0	9.13 m	5/2+	+1.93(7)	A	LRIS	1988AlO4	NP A477 37 (1988)
69 Tm 160	0	9.4 m	1-	+0.58(4)	A	LRIS	1988AlO4	NP A477 37 (1988)
69 Tm 161	0	30.2 m	7/2+	+2.90(7)	A	LRIS	1988AlO4	NP A477 37 (1988)
69 Tm 162	0	21.7 m	1-	+0.69(3)	A	LRIS	1988AlO4	NP A477 37 (1988)
69 Tm 164	0	2.0 m	1+	+0.71(5)	A	LRIS	1988AlO4	NP A477 37 (1988)
69 Tm 166	0	7.7 h	2+	+2.14(3)	A	LRIS	1988AlO4	NP A477 37 (1988)
69 Tm 168	0	93.1 d	3+	+3.23(7)	A	LRIS	1988AlO4	NP A477 37 (1988)
69 Tm 169	8	4.09 ns	3/2+	-1.2(1)	B	ME	1964Co08	PR 134 A94 (1964)
69 Tm 170	0	128.6 d	1+	+0.74(2)	A	LRIS	1988AlO4	NP A477 37 (1988)

Ytterbium

Muonic atom X-ray hyperfine structure

A. Assumes relation $Q(\text{spectroscopic}) = 2Q(\text{intrinsic})/7$ and $Q(\text{intrinsic}) 2+ (84 \text{ keV } 170\text{Yb} = 7.63(9) \text{ b.}$

70 Yb 155	0	1.79 s	(7/2-)	-0.5(3)	[173Yb]	LRIS	2000Ba16	PR C61 034304 (2000)
70 Yb 159	0	1.67 m	5/2(-)	-0.22(2)	[173Yb]	CLS	1983Ne13	HFI 15 181 (1983)
70 Yb 161	0	4.2 m	3/2-	+1.03(2)	[173Yb]	CLS	1983Ne13	HFI 15 181 (1983)
70 Yb 163	0	11.0 m	3/2-	+1.24(2)	[173Yb]	CLS	1983Ne13	HFI 15 181 (1983)
70 Yb 165	0	9.9 m	5/2-	+2.48(4)	[173Yb]	CLS	1983Ne13	HFI 15 181 (1983)
70 Yb 167	0	17.5 m	5/2-	+2.70(4)	[173Yb]	CLS	1983Ne13	HFI 15 181 (1983)
70 Yb 169	0	32.0 d	7/2+	+3.54(6)	[173Yb]	CLS	1983Ne13	HFI 15 181 (1983)
70 Yb 170	84	1.60 ns	2+	-2.18(3)	A		2001Ra27	ADNDT 78 1 (2001)
70 Yb 171	67	0.79 ns	3/2-	-2.34(7)	[170Yb 84 keV]	ME	1971PI03	NP A165 97 (1971)
	76	1.64 ns	5/2-	-2.22(7)	[170Yb 84 keV]	ME	1971PI03	NP A165 97 (1971)
70 Yb 172	79	1.65 ns	2+	-2.22(4)	[170Yb 84 keV]	ME	1971PI03	NP A165 97 (1971)
	260	0.122 ns	4+	-2.3(12)		CER	1970McZQ	ORNL-4513 56 (1970)
	1172	8.1 ns	3+	-2.9(3)	[170Yb 84 keV]	TDPAC	1970Wa25	ZP A238 35 (1970)
	1757	-	(1-)	-3.44(10)		Mu-X	1979Ho23	PR C20 1934 (1979)
	1822	-	(3-)	+1.97(10)		Mu-X	1979Ho23	PR C20 1934 (1979)

Reference isotope	70 Yb 173	0	stable	5/2-	+2.80(4)		Mu-X	1975Ze04	NP A254 315 (1975(
	70 Yb 174	77 253	1.79 ns 144 ps	2+ 4+	-2.18(5) -1.8(12)	[170Yb 84 keV]	ME CER	1971PI03 1970McZQ	NP A165 97 (1971) ORNL-4513 56 (1970)
	70 Yb 175	0	4.18 d	7/2-	+3.52(5)	[173Yb]	CLS	2012FI05	JPhys G39 125101 (2012)
	70 Yb 176	82 272 1050	1.76 ns 0.11 ns 11.4 s	2+ 4+ 8-	-2.28(6) -0.9(12) +5.30(8)	[170Yb 84 keV] [173Yb]	ME CER CLS	1967Ec01 1970McZQ 2007Bi14	PR 156 246 (1967) ORNL-4513 56 (1970) PL B645 330 (2007)
	70 Yb 177	0	1.91 h	9/2+	+4.03(6)	[173Yb]	CLS	2012FI05	JPhys G39 125101 (2012)
Lutetium	Muonic atom X-ray hyperfine structure								
	71 Lu 162	0	1.37 m	1-	+0.519(8)	[175Lu]	CLS	1998Ge13	Eur Phys J A3 225 (1998)
	71 Lu 164	0	3.14 m	1-	+0.608(7)	[175Lu]	CLS	1998Ge13	Eur Phys J A3 225 (1998)
	71 Lu 166	0 34	2.65 m 1.41 m	6- 3-	+4.33(4) +2.72(2)	[175Lu] [175Lu]	CLS CLS	1998Ge13 1998Ge13	Eur Phys J A3 225 (1998) Eur Phys J A3 225 (1998)
	71 Lu 167	0	51.5 m	7/2+	+3.28(2)	[175Lu]	CLS	1998Ge13	Eur Phys J A3 225 (1998)
	71 Lu 168	0 220	5.5 m 6.7 m	6- 3+	+4.77(6) +2.43(2)	[175Lu] [175Lu]	CLS CLS	1998Ge13 1998Ge13	Eur Phys J A3 225 (1998) Eur Phys J A3 225 (1998)
	71 Lu 169	0	34.1 h	7/2+	+3.48(3)	[175Lu]	CLS	1998Ge13	Eur Phys J A3 225 (1998)
	71 Lu 171	0	8.24 d	7/2+	+3.53(3)	[175Lu]	CLS	1998Ge13	Eur Phys J A3 225 (1998)
	71 Lu 172	0 42	6.70 d 3.7 m	4- 1-	+3.80(4) +0.76(3)	[175Lu] [175Lu]	CLS CLS	1998Ge13 1998Ge13	Eur Phys J A3 225 (1998) Eur Phys J A3 225 (1998)
	71 Lu 173	0	1.37 y	7/2+	+3.53(2)	[175Lu]	CLS	1998Ge13	Eur Phys J A3 225 (1998)
	71 Lu 174	0 171	3.31 y 142 d	1- 6-	+0.773(7) +4.80(5)	[175Lu] [175Lu]	CLS CLS	1998Ge13 1998Ge13	Eur Phys J A3 225 (1998) Eur Phys J A3 225 (1998)
Reference isotope	71 Lu 175	0	stable	7/2+	+3.49(2)		Mu-X	1979De29	NP A326 418 (1979)
	71 Lu 176	0 127	3.76 10 ¹⁰ y 3.66 h	7- 1-	+4.92(3) -1.450(12)	[175Lu] [175Lu]	A CLS	1985Br09 1998Ge13	NP A440 407 (1985) Eur Phys J A3 225 (1998)
	71 Lu 177	0 970	6.65 d 160 d	7/2+ 23/2	+3.39(3) +5.71(5)	[175Lu] [175Lu]	CLS CLS	1998Ge13 1998Ge13	Eur Phys J A3 225 (1998) Eur Phys J A3 225 (1998)
	71 Lu 178	0 120	28.4 m 23.1 m	1+ 9-	+0.708(10) +5.39(10)	[175Lu] [175Lu]	CLS CLS	1998Ge13 1998Ge13	Eur Phys J A3 225 (1998) Eur Phys J A3 225 (1998)
	71 Lu 179	0	4.59 h	7/2+	+3.32(3)	[175Lu]	CLS	1998Ge13	Eur Phys J A3 225 (1998)

Hafnium <i>Muonic atom X-ray hyperfine structure</i>									
	72 Hf 171	0	12.1 h	7/2+	+3.46(3)	[177Hf]	CLS	2000Ye02	J Phys G26 839 (2000)
	72 Hf 175	0	70 d	5/2-	+2.72(2)	[177Hf]	CLS	2002Ni12	PRL 88 094801 (2002)
	72 Hf 176	88	1.43 ns	2+	-2.10(2)	[177Hf]	Mu-X	1984Ta10	PR C30 350 (1984)
Reference isotope	72 Hf 177	0	stable	7/2-	+3.37(3)		Mu-X	1984Ta04	PR C29 1830 (1984)
		113	537 ps	9/2-	+1.30(2)	[177Hf]	Mu-X	1984Ta10	PR C30 350 (1984)
	72 Hf 178	93	1.49 ns	2+	-2.02(2)	[177Hf]	Mu-X	1984Ta10	PR C30 350 (1984)
		1147	4.0 s	23/2-	+4.99(4)	[177Hf]	CLS	2007Bi14	PL B645 330 (2007)
		2446	31 y	16+	+6.00(7)	[177Hf]	CLS	1994Bo15	PRL 72 2689 (1994)
Reference isotope	72 Hf 179	0	stable	9/2+	+3.79(3)		Mu-X	1984Ta04	PR C29 1830 (1984)
		123	37 ps	11/2+	+1.88(3)	[177Hf]	Mu-X	1984Ta10	PR C30 350 (1984)
	72 Hf 180	93	1.52 ns	2+	-2.00(2)	[177Hf]	Mu-X	1984Ta10	PR C30 350 (1984)
		1142	5.53 h	8-	+4.6(3)	[177Hf]	NO/S	1973Ka31	PL B46 62 (1973)
Tantalum <i>Pionic atom X-ray hyperfine structure</i>									
	73 Ta 171	184	46 ns	9/2-	(+) $3.1(2)$	[181Ta]	TDPAD	1995Do32	HFI 96 223 (1995)
	73 Ta 173	0	3.14 h	5/2-	-1.8(2)	[181Ta]	NO/S	1983Ed01	PL B133 44 (1983)
	73 Ta 175	0	10.5 h	7/2+	+3.5(3)	[181Ta]	NO/S	1983Ed01	PL B133 44 (1983)
	73 Ta 178	0 + x	9.3 m	1+	+0.63(6)	[181Ta]	NO/S	1983Ha49	HFI 15 105 (1983)
	73 Ta 179	0	1.82 y	7/2+	+3.27(4)	[181Ta]	CLS	1996Wa02	PR C53 611 (1996)
	73 Ta 180	75	$>7.1 \cdot 10^{15}$ y	9-	+4.80(3)	[181Ta]	CLS	1994Wa34	PR C50 4639 (1994)
Reference isotope	73 Ta 181	0	stable	7/2+	+3.17(2)		Pi-X	1983OI03	NP A 403 572 (1983)
		6	6.05 μ s	9/2-	+3.59(2)	[181Ta]	ME	1983Ei02	PL A93 259 (1983)
		482	10.8 ns	5/2+	+2.28(2)	[181Ta]	ME	1983Bu11	PL A97 217 (1983)
	73 Ta 182	0	115 d	3-	+2.6(3)	[181Ta]	NO/S	1991Fa12	PL A159 421 (1991)
Tungsten <i>There is no adopted reference efg for W A. Efg calculation in TI metal</i>									
	74 W 176	3746	41 ns	14+	6.0(8)	A	TDPAD	2002Io01	PL B541 219 (2002)
	74 W 179	3348	750 ns	35/2-	+3.9(10)	A	LEMS	2001Ba04	PRL 86 604 (2001)
	74 W 180	104	1.28 ns	2+	-2.1(4)	[182W 100 keV]	ME	1973Zi02	ZP 262 413 (1973)

	74 W 182	100	1.32 ns	2+	-2.1(4)		CER	1977RuZV	BAPS 22 1032 (1977)
	74 W 183	47 99	185 ps 0.73 ns	3/2- 5/2-	-1.8(4) -2.0(3)	[182W 100 keV] [182W 100 keV]	ME ME	1966Sh07 1967Ag02 1974Ge17	JPSJ 21 829 (1966) PR 155 1342 (1967) ZP 267 61 (1974)
	74 W 184	111	1.25 ns	2+	-1.9(2)	[182W 100 keV]	CER	1974Ge17 1977RuZV	ZP 267 61 (1974) BAPS 22 1032 (1977)
		904	1.80 ps	2+	+0.1(4)		CER	1977Ob02	NP A291 510 (1977)
	74 W 186	123 396 737	1.04 ns 36 ps 4.8 ps	2+ 4+ 2+	-1.6(3) -2.6(13) 1.3(3)	[182W 100 keV]	CER CER CER	1977RuZV 1970McZQ 1977Ob02	BAPS 22 1032 (1977) ORNL-4513 56 (1970) NP A291 510 (1977)
Rhenium	<i>Pionic atom X-ray hyperfine structure</i>								
	75 Re 182	0 0 + x	4.2 h 14.1 h	7+ 2+	+3.9(3) +1.7(2)	[185,187Re] [185,187Re]	NO/S/R NO/S/R	1983Ha49 1985Ha41 1981Er01	HFI 215 105 (1983) HFI 22 19 (1985) PR C23 1739 (1981)
	75 Re 183	0 496	70.0 d 7.8 ns	5/2+ 9/2-	+2.2(2) (+)3.5(4)	[185,187Re] [185,187Re]	NO/S/R TDPAC/R	1983Ha49 1978Ne14	HFI 215 105 (1983) HFI 4 211 (1978)
	75 Re 184	0	35.4 d	3-	+2.6(2)	[185,187Re]	NO/S/R	1983Ha49	HFI 215 105 (1983)
<i>Reference isotope</i>	75 Re 185	0	stable	5/2+	+2.07(5)		Mu-X	2020An04	PR C101 054313 (2020)
	75 Re 186	0	3.72 d	1-	+0.583(15)	[185,187Re]	AB/R	1981Bu13	ZP A302 290 (1981)
<i>Reference isotope</i>	75 Re 187	0 206	4.3 10 ¹⁰ y 555 ns	5/2+ 9/2-	+1.94(5) +2.87(7)	 [187Re]	Mu-X TDPAC/R	2020An04 1973Ha61	PR C101 054313 (2020) JCP 58 3339 (1973)
	75 Re 188	0	17.0 h	1-	+0.539(13)	[185,187Re]	AB/R	1981Bu13	ZP A302 290 (1981)
Osmium	<i>Muonic atom X-ray hyperfine structure</i>								
	76 Os 182	7049	150 ns	25+	4.2(2)	[188Os 155keV]	TDPAD	1991Br25	PL B264 17 (1991)
	76 Os 183	0	13.0 h	9/2+	+3.1(3)	[188Os 155keV]	NO/S	1985Ha41	HFI 22 19 (1985)
	76 Os 184	120	1.18 ns	2+	-2.7(12)	[188Os 155keV]	CER	1972La16	PR C6 613 (1972)
	76 Os 186	137	875 ps	2+	-1.63(4)		Mu-X	1981Ho22	PR C24 1667 (1981)
<i>Reference isotope</i>	76 Os 188	155 633 2121	688 ps 9.4 ps -	2+ 2+ (3-)	-1.46(4) +1.0(3) +1.69(9)	 [188Os 155keV]	Mu-X CER Mu-X	1981Ho22 1980Ba42 1979Ho23	PR C24 1667 (1981) PR C22 2383 (1980) PR C20 1934 (1979)
	76 Os 189	0 70	stable 1.62 ns	3/2- 5/2-	+0.86(3) -0.63(2)	[188Os 155keV] [188Os 155keV]	ME ME	1972Wa24 1972Wa24	ZP A254 112 (1972) ZP A254 112 (1972)

	76 Os 190	187 558	375 ps 12.8 ps	2+ 2+	-1.18(3) +0.8(5)	[188Os 155keV]	Mu-X CER	1981Ho22 1980Ba42	PR C24 1667 (1981) PR C22 2383 (1980)
	76 Os 191	0	15.4 d	9/2-	+2.53(16)	[188Os 155keV]	NO/S	1979Er09	NP A332 41 (1979)
	76 Os 192	206 489	288 ps 32.6 ps	2+ 2+	-0.96(3) -0.7(3)	[188Os 155keV]	Mu-X CER	1981Ho22 1980Ba42	PR C24 1667 (1981) PR C22 2383 (1980)
	76 Os 193	0	30.1 h	3/2-	+0.48(6)	[188Os 155keV]	NO/S,R	1985Be03 1979Er09	JPhys G11 287 (1985) NP A332 41 (1979)
Iridium	<i>Muonic atom X-ray hyperfine structure</i>								
	<i>A. Estimated efg at Ir in hcp Co metal crystal</i>								
	<i>B. Estimated efg at Ir in Os metal polycrystal</i>								
	77 Ir 182	0	15 m	3+	-1.7(6)	[191Ir]	RIMS/LS	2006Ve10	Eur Phys J A30 489 (2006)
	77 Ir 183	0	58 m	5/2-	-1.8(7)	[191Ir]	RIMS/LS	2006Ve10	Eur Phys J A30 489 (2006)
	77 Ir 184	0	3.09 h	5-	+2.41(3)	A	QI-NMR/ON	1996Se15	PRL 77 5016 (1996)
	77 Ir 185	0	14.4 h	5/2-	-1.84(12)	A	NMR/ON R	1988Oh02	J Phys G 14 365 (1988)
	77 Ir 186	0 0+x	16.64 h 1.90 h	5+ 2(-)	-2.55(3) +1.456(17)	A A	QI-NMR/ON QI-NMR/ON	1996Se15 1996Se15	PRL 77 5016 (1996) PRL 77 5016 (1996)
	77 Ir 187	0 434	10.5 h 152 ns	3/2+ 11/2-	+0.941(11) 2.33(14)	A [193Ir]	QI-NMR/ON TDPAC	1996Se15 1978HaXO	PRL 77 5016 (1996) ARHMI 52 1977
	77 Ir 188	0	41.5 h	1(-)	+0.484(6)	A	QI-NMR/ON	1996Se15	PRL 77 5016 (1996)
	77 Ir 189	0	13.2 d	3/2+	+0.82(8) [+0.878(10)]	[191Ir]	RIMS/LS Estimated	2006Ve10 1996Se15	Eur Phys J A30 489 (2006) PRL 77 5016 (1996)
	77 Ir 190	0	11.8 d	(4)+	+2.87(16)	A	NO/S	1980Mu07	HFI 7 481 (1980)
<i>Reference isotope</i>	77 Ir 191	0	stable	3/2+	+0.816(9)		Mu-X	1984Ta04	PR C29 1830 (1984)
	77 Ir 192	0	73.8 d	4-	+2.15(6)	A	QI-NMR/ON	1996Se15	PRL 77 5016 (1996)
	77 Ir 193	0	stable	3/2+	+0.751(9)		Mu-X	1984Ta04	PR C29 1830 (1984)
	77 Ir 194	0	19.3 h	1-	+0.339(12)	[191Ir]	NMR/ON	1985Ed02	PR C32 582 (1985)
Platinum	<i>There is no adopted reference efg for Pt.</i>								
	<i>A. For details of the efg used see 1992Hi07</i>								
	<i>B. Estimated efg at Pt in osmium metal</i>								
	78 Pt 183	35	43 s	7/2-	+3.4(3)	A	LS	1999Le52 1992Hi07	PR C60 054310 (1999) ZP A342 1 (1992)

	78 Pt 185	0	70.9 m	9/2+	+3.73(17)	A	LS	1999Le52 1992Hi07	PR C60 054310 (1999) ZP A342 1 (1992)
	78 Pt 187	0	2.35 h	3/2-	-1.02(4)	A	RIMS/LS	1992Hi07 1989Du01	ZP A342 1 (1992) PL B217 401 (1989)
	78 Pt 189	0	10.9 h	3/2-	-0.95(4)	A	RIMS/LS	1992Hi07 1989Du01	ZP A342 1 (1992) PL B217 401 (1989)
	78 Pt 191	0	2.83 d	3/2-	-0.87(4)	A	RIMS/LS	1992Hi07 1989Du01	ZP A342 1 (1992) PL B217 401 (1989)
	78 Pt 192	317	43.7 ps	2+	+0.6(2)		CER	1987Gy01	NP A470 415 (1987)
	78 Pt 194	328	41.8 ps	2+	+0.48(14)		CER	1986Gy04	NP A458 165 (1986)
	78 Pt 195	259	4.01 d	13/2+	+1.4(6)	B	NO/S	1985Ed05	PL B158 371 (1985)
	78 Pt 196	356 689 877 1526	34 ps 33.8 ps 3.55 ps 0.98 ps	2+ 2+ 4+ 6+	+0.62(8) -0.39(16) +1.03(12) -0.2(3)		CER CER CER CER	1992Li14 1992Li14 1992Li14 1992Li14	NP A548 308 (1992) NP A548 308 (1992) NP A548 308 (1992) NP A548 308 (1992)
	78 Pt 198	407	22.3 ps	2+	+0.42(12)		CER	1986Gy04	NP A458 165 (1986)
Gold	<i>Muonic atom X-ray hyperfine structure</i>								
	79 Au 184	0	20.6 s 47.6 s	5 2	+4.7(3) +1.90(16)	[197Au] [197Au]	CLS CLS	1997Le22 1997Le22	PRL 79 2213 (1997) PRL 79 2213 (1997)
	79 Au 185	0	4.25 m	5/2-	-1.10(10)	[186Au, 197Au]	CLS	1992Ki30 1994Pa37	NIMPR B70 537 (1992) NP A580 173 (1994)
	79 Au 186	0	10.7 m	3-	+3.10(6)	[186Au, 197Au]	CLS	1992Ki30 1994Pa37	NIMPR B70 537 (1992) NP A580 173 (1994)
	79 Au 191	0	3.18 h	3/2+	+0.72(2)	[197Au]	CLS	1994Pa37	NP A580 173 (1994)
	79 Au 192	0	4.94 h	1-	-0.228(8)	[197Au]	CLS	1994Pa37	NP A580 173 (1994)
	79 Au 193	0 290	17.65 h 3.9 s	3/2+ 11/2-	+0.66(2) +1.98(6)	[197Au] [197Au]	CLS MAPON	1994Pa37 1996Se06	NP A580 173 (1994) NP A602 41 (1996)
	79 Au 194	0	38.0 h	1-	-0.240(9)	[197Au]	CLS	1994Pa37	NP A580 173 (1994)
	79 Au 195	0 319	186 d 30.5 s	3/2+ 11/2-	+0.607(18) +1.87(6)	[197Au] [197Au]	QI-NMR/ON MAPON	1993Hi10 1996Se06	NP A562 205 (1993) NP A602 41 (1996)
	79 Au 196	0	6.17 d	2-	+0.81(7)	[197Au]	NMR/ON	1987Oh11	PR C36 2072 (1987)
<i>Reference isotope</i>	79 Au 197	0	stable	3/2+	+0.547(16)		Mu-X	1974Po02	NP A230 413 (1974)

	409	7.73 s	11/2-	+1.68(5)	[197Au]	MAPON	1996Se06	NP A602 41 (1996)
79 Au 198	0	2.694 d	2-	+0.640(19)	[197Au]	NMR/ON	1993Hi10	NP A562 205 (1993)
79 Au 199	0	3.14 d	3/2+	+0.510(16)	[197Au]	NMR/ON	1993Hi10	NP A562 205 (1993)
Mercury	<i>Efg calculations in the 3P1 state of neutral Hg</i>							
80 Hg 185	99	21.6 s	13/2+	+0.2(3)	[201Hg]	β-RADOP	1979Da06	PL B82 199 (1979)
80 Hg 187	0	1.9 m	3/2-	-0.75(18)	[201Hg]	β-RADOP	1986UI02	ZP A325 247(1986)
	0 + x	2.4 m	13/2+	+0.5(3)	[201Hg]	β-RADOP	1979Da06	PL B82 199 (1979)
80 Hg 188	2724	134 ns	12+	0.91(11)	[199Hg 158 keV]	TDPAD	1984Dr09	PL B149 311 (1984)
80 Hg 189	0	7.6 m	3/2-	-0.8(3)	[201Hg]	β-RADOP	1986UI02	ZP A325 247(1986)
	0 + x	8.6 m	13/2+	+0.66(19)	[201Hg]	β-RADOP	1979Da06	PL B82 199 (1979)
80 Hg 190	2621	23 ns	12+	1.17(14)	[199Hg 158 keV]	TDPAD	1984Dr09	PL B149 311 (1984)
80 Hg 191	0	49 m	3/2-	-0.80(13)	[201Hg]	β-RADOP	1986UI02	ZP A325 247(1986)
	140	50.8 m	13/2+	+0.6(2)	[201Hg]	β-RADOP	1979Da06	PL B82 199 (1979)
80 Hg 193	0	3.80 h	3/2-	-0.7(3)	[201Hg]	O	1974Fu06	PR A9 593 (1974)
	141	11.8 h	13/2+	+0.92(2)	[201Hg]	O	1966Da07	PR 147 861 (1966)
							1974Re05	PR A9 1776 (1974)
80 Hg 195	176	41.6 h	13/2+	+1.08(2)	[201Hg]	O	1965Sm01	PR A137 330 (1965)
80 Hg 197	134	8.1 ns	5/2-	+0.081(6)	[199Hg 158 keV]	TDPAC	1980He05	NP A337 261 (1980)
	299	23.8 h	13/2+	+1.25(3)	[201Hg]	O	1961Br17	J Phys Radium 22 412 (1961)
80 Hg 198	412	23 ps	2+	+0.68(12) or +0.84(12)		CER	1979Bo16	ZP A291 245 (1979)
							1984Fe08	NP A425 373 (1984)
80 Hg 199	158	2.45 ns	5/2-	+0.95(7)		Mu-X	1979Ha08	NP A314 361 (1979)
	208	69 ps	3/2-	+0.62(15)		Mu-X	1979Ha08	NP A314 361 (1979)
	532	42.6 m	13/2+	+1.2(3)	[201Hg]	β-RADOP	1979Da06	PL B82 199 (1979)
80 Hg 200	368	46.4 ps	2+	+0.96(11) or +1.11(11)		CER	1979Bo16	ZP A291 245 (1979)
<i>Reference isotope</i> 80 Hg 201	0	stable	3/2-	+0.387(6)			2005Bi03	PR A71 012502 (2005)
							1961Ko05	PR 121 1104 (1961)
80 Hg 202	440	27.3 ps	2+	+0.87(13) or +1.01(13)		CER	1980Sp05	NP A345 252 (1980)
80 Hg 203	0	46.6 d	5/2-	+0.344(7)	[201Hg]	O	1970Re14	PR A2 1135 (1970)

80 Hg 204	437	40.3 ps	2+	+0.4(2)		CER	1981Es03	NP A362 227 (1981)
80 Hg 206	2102	2.15 μ s	5-	0.74(15)	[199Hg 158 keV]	TDPAD	1984Ma43	PR C30 1702 (1984)
Thallium								
<i>There is no adopted reference efg for Tl.</i>								
<i>A. For reference to the efg used in CLS studies see 1987Bo44 (PR C36 2560 (1987))</i>								
<i>B. Estimated efg in In metal</i>								
81 Tl 187	335	15.6 s	(9/2-)	-2.43(5)	A	CLS	1993ScZW	IoP Conf 132 221 (1993)
81 Tl 188	0 + x	71 s	7+	+0.129(4)	A	CLS	1992Me07	ZP A341 475 (1992)
81 Tl 189	281	1.4 m	9/2-	-2.29(4)	A	CLS	1987Bo44	PR C36 2560 (1987)
81 Tl 190	0 + x	2.6 m	2-	-0.329(9)	A	CLS	1992Me07	ZP A341 475 (1992)
	0 + y	3.7 m	7+	+0.285(14)	A	CLS	1992Me07	ZP A341 475 (1992)
81 Tl 191	299	5.2 m	9/2-	-2.23(2)	A	CLS	1992Me07	ZP A341 475 (1992)
81 Tl 192	0 + x	9.6 m	2-	-0.328(11)	A	CLS	1992Me07	ZP A341 475 (1992)
	0 + y	10.8 m	7+	+0.46(2)	A	CLS	1992Me07	ZP A341 475 (1992)
	251 + x	296 ns	8-	0.44(7)	B	TDPAD	1982Sc27	ZP B49 23 (1982)
81 Tl 193	365	2.11m	9/2-	-2.20(2)	A	CLS	1987Bo44	PR C36 2560 (1987)
81 Tl 194	0	33.0 m	2-	-0.282(7)	A	CLS	1992Me07	ZP A341 475 (1992)
	0 + y	32.8 m	7+	+0.607(16)	A	CLS	1992Me07	ZP A341 475 (1992)
81 Tl 196	0	1.84 h	2-	-0.178(14)	A	CLS	1992Me07	ZP A341 475 (1992)
	394	1.41 h	7+	+0.76(2)	A	CLS	1992Me07	ZP A341 475 (1992)
81 Tl 205	204	1.46 ns	3/2+	+0.74(15)		Mu-X	1972Ch07	NP A181 25 (1972)
	2623	-	(5/2)-	-0.5(2)		Mu-X	1972Ch07	NP A181 25 (1972)
Lead								
<i>Efg in 3P1 state of neutral Pb</i>								
<i>A. Efg in 1D2 state in neutral Pb</i>								
<i>B. Normalised to estimated Q of 206Pb 4027 keV</i>								
<i>C. Obtained from theory of relaxation in Hg metal</i>								
82 Pb 191	138	2.18 m	13/2+	+0.085(5)	A	CLS	1991Du07	ZP A341 39 (1991)
82 Pb 192	2581+d	1.09 μ s	12+	0.32(4)	B	TDPAD	2007Io03	PL B650 141 (2007)
	2743	756 ns	11-	2.9(3)	B	TDPAD	2007Io03	PL B650 141 (2007)
82 Pb 193	100	5.8 m	13/2+	+0.195(10)	A	CLS	1991Du07	ZP A341 39 (1991)
	1586 + x	20.5 ns	(21/2-)	0.22(2)	B	TDPAD	2004Ba31	Eur Phys J A20 191 (2004)
	2585 + x	9.4 ns	(27/2-)	2.6(3)	B	TDPAD	2011Ba02	PR C83 014304 (2011)
			(29/2-)	2.8(3)	B	TDPAD	2004Ba31	Eur Phys J A20 191 (2004)
	2613 + x	180 ns	(33/2+)	0.45(4)	B	TDPAD	2004Ba31	Eur Phys J A20 191 (2004)
82 Pb 194	2628	370 ns	(12+)	0.49(3)	B	TDPAD	1985St16	ZP A322 83 (1985)

	2933	133 ns	(11)-	3.6(4)	B	TDPAD	2007Io03	PL B650 141 (2007)	
82 Pb 195	203	15.0 m	13/2+	+0.306(15)	A	CLS	1991Du07	ZP A341 39 (1991)	
82 Pb 196	2694	270 ns	12+	0.65(5)	B	TDPAD	1981Zy02	HFI 9 109 (1981)	
	3193	72 ns	11-	(-)-3.4(7)	B	LEMS	2002Vy01	PRL 88 102502 (2002)	
82 Pb 197	0	8.1 m	3/2-	-0.08(17)		CLS	1986An06	ZP A451 471 (1986)	
	319	42.9 m	13/2+	+0.378(19)	A	CLS	1991Du07	ZP A341 39 (1991)	
82 Pb 198	2822	212 ns	(12)+	0.75(5)	B	TDPAD	1981Zy02	HFI 9 109 (1981)	
82 Pb 199	0	1.5 h	3/2-	'+0.08(9)		CLS	1986An06	ZP A451 471 (1986)	
82 Pb 200	2154	45.1 ns	(7-)	0.32(2)	B	TDPAD	1979MaYq	AECL-6680 27 (1979)	
	2183	448 ns	(9-)	0.40(2)	B	TDPAD	1979MaYq	AECL-6680 27 (1979)	
	3006	199 ns	(12+	0.79(3)	B	TDPAD	1979Ma37	PL B88 48 (1979)	
82 Pb 201	0	9.33 h	5/2-	0.01(4)		CLS	1986An06	ZP A451 471 (1986)	
	2720	63 ns	25/2-	0.46(2)	B	TDPAD	1979MaYq	AECL-6680 27 (1979)	
82 Pb 202	2170	3.54 h	9-	+0.58(9)		CLS	1986An06	ZP A451 471 (1986)	
	2208	65 ns	7-	0.28(2)	B	TDPAD	1979MaYq	AECL-6680 27 (1979)	
82 Pb 203	0	51.9 h	5/2-	+0.10(5)		CLS	1986An06	ZP A451 471 (1986)	
	1922	42 ns	21/2+	0.85(3)	B	TDPAD	1979MaYq	AECL-6680 27 (1979)	
82 Pb 204	899	2.88 ps	2+	+0.23(9)		CER	1978Jo04	PL B72 307 (1978)	
	1274	265 ns	4+	0.44(2)	B	TDPAD	1979MaYq	AECL-6680 27 (1979)	
82 Pb 205	0	1.73 10 ⁷ y	5/2-	+0.23(4)		CLS	1986An06	ZP A451 471 (1986)	
	1014	5.55 ms	13/2+	0.30(5)	C	QIR	1974Ri03	PS 11 228 (1975)	
	3196	217 ns	25/2-	0.63(3)	B	TDPAD	1979MaYq	AECL-6680 27 (1979)	
82 Pb 206	803	8.3 ps	2+	+0.05(9)		CER	1978Jo04	PL B72 307 (1978)	
	2200	125 μs	7-	0.33(5)	C	QIR	1974Ri03	PS 11 228 (1975)	
	4027	202 ns	12+	[0.51(2)]	from B(E2)	not measured	1979Ma37	PL B88 48 (1979)	
82 Pb 208	2615	16.7 ps	3-	-0.34(15)		CER	1984Ve07	AuJP 37 123 (1984)	
	4086	0.80fs	2+	-0.7(3)		CER	1984Ve07	AuJP 37 123 (1984)	
Reference isotope	82 Pb 209	0	3.23 h	9/2+	-0.27(17)		CLS	1986An06	ZP A451 471 (1986)
	82 Pb 211	0	36.1 m	9/2+	+0.09(6)		CLS	1986An06	ZP A451 471 (1986)
Bismuth	Efg calculations in the 4P3/2 state of neutral Bi						2001Bi23	PRL 87 133003 (2001)	
83 Bi 193	0	63.6 s	(9/2-)	-2.3(5)	[209Bi]	ISLS	2016Ba42	PR C94 024334 (2016)	
83 Bi 195	0	183 s	(9/2-)	-1.7(6)	[209Bi]	ISLS	2016Ba42	PR C94 024334 (2016)	

	83 Bi 197	0	9.33 m	(9/2-)	-1.7(10)	[209Bi]	ISLS	2016Ba42	PR C94 024334 (2016)
	83 Bi 202	0	1.71 h	[5+]	-1.00(9)	[209Bi]	LFRS	1996Ca02 2001Bi23	NP A598 61 (1996) PRL 87 133003 (2001)
		605 + x	3.04 μ s	[6+]	-1.21(9)	[209Bi]	LFRS	1996Ca02 2001Bi23	NP A598 61 (1996) PRL 87 133003 (2001)
		2597 + x	310 ns	10- 17+	0.14(2) 0.45(2)	[209Bi] [209Bi]	TDPAD TDPAD	1987Ma65 1987Ma65	HFI 34 47 (1987) HFI 34 47 (1987)
	83 Bi 203	0	11.8 h	9/2-	-0.93(7)	[209Bi]	LFRS	1996Ca02 2001Bi23	NP A598 61 (1996) PRL 87 133003 (2001)
	83 Bi 204	0	11.22 h	6+	-0.68(20)	[209Bi]	LFRS	1996Ca02 2001Bi23	NP A598 61 (1996) PRL 87 133003 (2001)
		806	13.0 ms	10-	0.074(2)	[209Bi]	LEMS	1991Sc14	PR C43 2560 (1991)
	83 Bi 205	0	15.3 d	9/2-	-0.81(3)	[209Bi]	LRFS	2000Pe30 2001Bi23	J Phys G26 1829(2000) PRL 87 133003 (2001)
	83 Bi 206	0 1045	6.243 d 0.89 ms	6+ (10-)	-0.54(4) 0.057(11)	[209Bi] [209Bi]	LRFS LEMS	2000Pe30 2001Bi23 1991Sc14	J Phys G26 1829(2000) PRL 87 133003 (2001) PR C43 2560 (1991)
	83 Bi 207	0 2101	31.6 γ 182 μ s	9/2- 21/2+	-0.76(2) 0.051(9)	[209Bi] [209Bi]	LRFS LEMS	2000Pe30 2001Bi23 1991Sc14	J Phys G26 1829(2000) PRL 87 133003 (2001) PR C43 2560 (1991)
	83 Bi 208	0	3.7 10 ⁵ γ	5+	-0.70(8)	[209Bi]	LRFS	2000Pe30 2001Bi23	J Phys G26 1829(2000) PRL 87 133003 (2001)
Reference isotope	83 Bi 209	0	2.0 10 ¹⁹ γ	9/2-	-0.516(15)		AB	1970Hu05 2001Bi23	PR A1 685 (1970) PRL 87 133003 (2001)
		2564	15 fs	(9/2)+	+0.15(7)	[209Bi]	Mu-X	1972Le07	NP A180 14 (1972)
		2741	9.1 ps	15/2+	0.0(5)	[209Bi]	Mu-X	1972Le07	NP A180 14 (1972)
	83 Bi 210	0	5.01 d	1-	+0.190(6)	[209Bi]	AB	1962Al02 2001Bi23	PR 125 256 (1962) PRL 87 133003 (2001)
		271	3.0 10 ⁶ γ	9-	-0.66(7)	[209Bi]	LRFS	2000Pe30 2001Bi23	J Phys G26 1829(2000) PRL 87 133003 (2001)
	83 Bi 211	0	2.14 m	9/2-	-0.7(3)	[213Br]	LRS	2018Ba03	PR C97 014322 (2018)
	83 Bi 212	0	60.6 m	1(-)	+0.1(4)	[209Bi]	LRFS	2000Pe30 2001Bi23	J Phys G26 1829(2000) PRL 87 133003 (2001)
	83 Bi 213	0	45.6 m	9/2-	-0.83(5)	[209Bi]	LRFS	2000Pe30 2001Bi23	J Phys G26 1829(2000) PRL 87 133003 (2001)

Polonium

There is no adopted reference efg for Po.

A. Calculated efg in 5S2 level of Po atom [1991Ko32][1961Ol01] with estimated Sternheimer correction. Ratios to 207Po B value.

B. The moments quoted are based on a calculated value for the 1557 keV, 8+, state in 210Po [1991Be03, NPA522 483 (1991)].

Reference isotope	84 Po 193	0 95	388 ms 245 ms	(3/2-) (13/2+)	-1.3(3) +1.1(5)	[207Po] [207Po]	LRIS LRIS	2014Se07 2014Se07	PR C89 034323 (2014) PR C89 034323 (2014)
	84 Po 195	0 ~ 230	4.64 s 1.92 s	(3/2-) (13/2+)	-0.9(3) +1.3(5)	[207Po] [207Po]	LRIS LRIS	2014Se07 2014Se07	PR C89 034323 (2014) PR C89 034323 (2014)
	84 Po 197	0 204	84 s 32 s	(3/2-) (13/2+)	-0.4(2) +1.3(5)	[207Po] [207Po]	LRIS LRIS	2014Se07 2014Se07	PR C89 034323 (2014) PR C89 034323 (2014)
	84 Po 198	605	~	2+	+2.1(11)		CER	2015Ke06	PR C92 054301 (2015)
	84 Po 199	0 310	5.47 m 4.17 m	(3/2-) (13/2+)	-0.27(15) +1.4(4)	[207Po] [207Po]	LRIS LRIS	2014Se07 2014Se07	PR C89 034323 (2014) PR C89 034323 (2014)
	84 Po 200	666 1774	~ 61 ns	2+ 8+	+0.05(15) (-)1.38(7)		CER TDPAD	2015Ke06 1987Ma65	PR C92 054301 (2015) HFI 34 47 (1987)
	84 Po 201	0 424	15.6 m 9.0 m	3/2- 13/2+	+0.1(1) +1.3(4)	[207Po] [207Po]	LRIS LRIS	2014Se07 2014Se07	PR C89 034323 (2014) PR C89 034323 (2014)
	84 Po 202	677 1712	~ 110 ns	2+ 8+	-0.5(9) (-)1.21(16)		CER LEMS	2015Ke06 1997Ne06	PR C92 054301 (2015) NP A625 668 (1997)
	84 Po 203	0 642	36.7 m 45 s	5/2- 13/2+	+0.17(10) +1.2(2)	[207Po] [207Po]	LRIS LRIS	2014Se07 2014Se07	PR C89 034323 (2014) PR C89 034323 (2014)
	84 Po 204	1639	154 ns	8+	(-)1.14(5)	B	TDPAD	1987Ma65	HFI 34 47 (1987)
	84 Po 206	1586	232 ns	8+	(-)1.02(4)	B	TDPAD	1987Ma65	HFI 34 47 (1987)
	84 Po 207	0	5.80 h	5/2-	+0.28(3)	A	LRIS	2014Se07	PR C89 034323 (2014)
	84 Po 208	1528	350 ns	8+	(-)0.90(4)	B	TDPAD	1987Ma65	HFI 34 47 (1987)
	84 Po 209	1473	89 ns	17/2-	(-)0.39(8)	B	TDPAD	1983Da01	NP A394 245 (1983)
	84 Po 210	1557 2849 4372 5058	99 ns 19.6 ns 54 ns 263 ns	8+ 11- 13- 16+	[-0.55(2)] (-)0.86(11) (-)0.90(7) (-)1.30(2)	from B(E2) B B B	not measured TDPAD TDPAD TDPAD	1991Be03 1991Be03 1991Be03 1991Be03	NP A522 483 (1991) NP A522 483 (1991) NP A522 483 (1991) NP A522 483 (1991)
	84 Po 211	0	0.516 s	9/2+	-0.77(15)	[207Po]	LRIS	2014Se07	PR C89 034323 (2014)

Astitine

There is no adopted reference efg for At.

A. The moments quoted are based on a calculated value for the 1417 keV, 21/2-, state in 211At [1995Ba66 NP A591 104 (1995)].

B. Calculated efg in atomic ground state of At. (2018Cu02 PR C97 054327 (2018))

85 At 195	33	143 ms	(7/2-)	-2.04(10)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 196	0	387 ms	(3+)	-0.6(4)	B	LRIS	2018Cu02	PR C97 054327 (2018)

85 At 197	0	381 ms	(9/2-)	-1.2(6)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 198	0	4.1 s	(3+)	-0.6(3)	B	LRIS	2018Cu02	PR C97 054327 (2018)
	0 + x	1.03 s	(10+)	+0.4(3)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 199	0	6.92 s	(9/2-)	-1.0(5)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 200	0	43 s	(3+)	-0.5(5)	B	LRIS	2018Cu02	PR C97 054327 (2018)
	112	47 s	(7+)	-1.0(5)	B	LRIS	2018Cu02	PR C97 054327 (2018)
	344	3.5 s	(10-)	+0.5(4)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 201	0	83 s	(9/2-)	-1.0(5)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 202	0	184 s	(3+)	-0.5(3)	B	LRIS	2018Cu02	PR C97 054327 (2018)
	0 + x	182 s	(7+)	-0.7(3)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 203	0	7.4 m	(9/2-)	-0.7(4)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 204	0	9.1 m	7+	-0.6(3)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 205	0	26.9 m	9/2-	-0.6(3)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 206	0	30.6 m	(6+)	-0.4(2)	B	LRIS	2018Cu02	PR C97 054327 (2018)
			(5+)	-0.3(2)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 207	0	1.81 h	9/2-	-0.5(3)	B	LRIS	2018Cu02	PR C97 054327 (2018)
85 At 208	0	1.63 h	6+	-0.4(3)	B	LRIS	2018Cu02	PR C97 054327 (2018)
	1090	48 ns	10-	(-)1.67(18)	A	LEMS	1991Sc15	PR C43 2566 (1991)
85 At 209	0	5.42 h	9/2-	-0.4(2)	B	LRIS	2018Cu02	PR C97 054327 (2018)
	1428	26 ns	21/2-	(-)0.78(6)	A	TDPAD	1983Ma08	PL B122 27 (1983)
	2429	890 ns	29/2+	(-)1.49(9)	A	TDPAD	1983Ma08	PL B122 27 (1983)
85 At 210	0	8.1 h	(5)+	-0.4(2)	B	LRIS	2018Cu02	PR C97 054327 (2018)
	1363	24.8 ns	11+	(-)0.64(5)	A	TDPAD	1983Ma08	PL B122 27 (1983)
	2550	482 ns	15-	(-)1.21(7)	A	TDPAD	1983Ma08	PL B122 27 (1983)
	4028	5.66 μ s	19+	(-)2.16(18)	A	LEMS	1991Sc15	PR C43 2566 (1991)
85 At 211	1417	35.1 ns	21/2-	[(-)0.524(10)]	from B(E2)	not measured	1995Ba66	NP A591 104 (1996)
	2641	50.8 ns	29/2+	(-)1.01(7)	A	TDPAD	1983Ma08	PL B122 27 (1983)
	4815	4.2 μ s	39/2-	(-)1.88(19)	A	LEMS	1991Sc15	PR C43 2566 (1991)
85 At 217	0	32.3 ms	(9/2-)	-1.40(8)	B	ISLS	2019Ba22	PR C99 054317 (2019)
85 At 218	0	1.5 s	(2,3)	+0.6(4)	B	ISLS	2019Ba22	PR C99 054317 (2019)
85 At 219	0	56 s	(9/2-)	-1.17(6)	B	ISLS	2019Ba22	PR C99 054317 (2019)

Radon	Estimated efg in Rn atom							[CERN EP/87 51 (1987)]	
	A. Normalised to Q of 1694 keV, 8+ state in 212Rn estimated from B(E2).								
	86 Rn 203	362	26.9 s	(13/2+)	+1.28(13)	209Rn	CLS	1987OtZW	CERN EP/87 51 (1987)
	86 Rn 205	0	2.83 m	5/2-	+0.062(6)	209Rn	CLS	1987OtZW	CERN EP/87 51 (1987)
	86 Rn 207	0	9.3 m	5/2-	+0.22(2)	209Rn	CLS	1987OtZW	CERN EP/87 51 (1987)
	86 Rn 208	1828	487 ns	8+	0.41(5)	A	TDPAD	1986Be40	PL B182 11 (1986)
	86 Rn 209	0	28.8 m	5/2-	+0.31(3)		efg in Rn atom	1987OtZW	CERN EP/87 51 (1987)
	86 Rn 210	1665+x 3812+x	644 ns 1.06 μs	(8+) (17)-	0.32(4) 0.89(10)	A A	TDPAD TDPAD	1986Be40 1986Be40	PL B182 11 (1986) PL B182 11 (1986)
	86 Rn 211	1578+x 8856 + y	596 ns 201 ns	17/2- 63/2-	0.19(2) 1.6(2)	A A	TDPAD TDPAD	1985Da14 1985Da14	PRL 55 1269 (1985) PRL 55 1269 (1985)
	86 Rn 212	1694	0.91 ms	8+	[-0.18(2)]	from B(E2)	not measured	1985Da13	NP A441 501 (1985)
	86 Rn 219	0	3.96 s	5/2+	+1.15(12)	209Rn	CLS	1987OtZW	CERN EP/87 51 (1987)
	86 Rn 221	0	25 m	7/2+	-0.47(5)	209Rn	CLS	1987OtZW	CERN EP/87 51 (1987)
	86 Rn 223	0	24.3 m	7/2	+0.80(8)	209Rn	CLS	1988NeZZ	Bk88 NFFS 126 (1988)
	86 Rn 225	0	4.66 m	7/2-	+0.84(8)	209Rn	CLS	1988NeZZ	Bk88 NFFS 126 (1988)
Francium	A. Efg calculated in the 7p 2P3/2 state of the Fr atom.							PR A92 052506(2015)	
	B. Efg calculated in the 8p 2P3/2 state of the Fr atom.							PR A92 052506(2015)	
	C. Normalised to calculated Q of the 2538 keV 29/2+ state in 211Fr.								
	87 Fr 203	0	0.55 s	(9/2-)	-0.47(2)	B	CLS	2017Wi11	PR C96 034317 (2017)
	87 Fr 204	0	1.8 s	(3+)	-0.141(9)	A	CLS	2015Vo05	PR C91 044307 (2015)
		41	1.6 s	(7+)	-0.613(17)	A	CLS	2015Vo05	PR C91 044307 (2015)
		316	0.8 s	(10+)	+1.39(3)	A	CLS	2015Vo05	PR C91 044307 (2015)
	87 Fr 205	0	3.9 s	(9/2-)	-0.305(18)	B	CLS	2016Ly01	PR C93 014319 (2016)
	87 Fr 206	0	16 s	3+	-0.354(9)	B	CLS	2016Ly01	PR C93 014319 (2016)
		95 - x	16 s	7(+)	-0.143(11)	B	CLS	2016Ly01	PR C93 014319 (2016)
		672 - x	0.7 s	10(-)	+1.26(3)	B	CLS	2016Ly01	PR C93 014319 (2016)
	87 Fr 207	0	14.8 s	9/2-	-0.24(2)	B	CLS	2017Wi11	PR C96 034317 (2017)
	87 Fr 208	0	59.1	7+	+0.052(11)	A	CLS	2015Vo05	PR C91 044307 (2015)
	87 Fr 209	0	50.5 s	9/2-	-0.26(2)	A	ABLS	1985Co24	PL B163 66 (1985)

87 Fr 210	0	3.18 m	6+	+0.21(2)	A	ABLS	1985Co24	PL B163 66 (1985)
87 Fr 211	0	3 10 m	9/2-	-0.21(2)	A	ABLS	1985Co24	PL B163 66 (1985)
	2423	146 ns	29/2+	(-)1.07(18)	C	LEMS	1991Ha02	PR C43 514 (1991)
	4657	123 ns	45/2-	(-)2.0(6)	C	LEMS	1991Ha02	PR C43 514 (1991)
87 Fr 212	0	20.0 m	5+	-0.100(2)	A	CLS	1987Du13	Europhys Lett 3 175 (1987)
	2492	604 ns	(15-)	(-)0.84(13)	C	TDPAD	1990By03	NP A516 145 (1990)
	5854	312 ns	(27-)	(-)1.7(3)	C	TDPAD	1990By03	NP A516 145 (1990)
87 Fr 213	0	34.8 s	9/2-	-0.138(3)	A	CLS	1987Du13	Europhys Lett 3 175 (1987)
	2538	238 ns	29/2+	[-0.70(7)]	calculated	not measured	1990By03	NP A516 145 (1990)
	8095	3.1 μ s	65/2-	(-)2.2(5)	C	LEMS	1991Ha02	PR C43 514 (1991)
87 Fr 214	638	103 ns	11+	0.8(2)	C	LEMS	1995Ne06	PR C51 3483 (1995)
	6447 + y	108 ns	32+ or 33+	2.2(5)	C	LEMS	1995Ne06	PR C51 3483 (1995)
87 Fr 219	0	22.5 ms	5/2-	-1.19(2)	A	CLS	2015De28	PRL 115 132501 (2015)
87 Fr 220	0	27.4 s	1+	+0.487(8)	A	CLS	1987Du13	Europhys Lett 3 175 (1987)
87 Fr 221	0	4.9 m	5/2-	-1.02(3)	A	CLS	2015De28	PRL 115 132501 (2015)
87 Fr 222	0	14.2 m	2-	+0.51(4)	A	ABLS	1985Co24	PL B163 66 (1985)
87 Fr 223	0	22.0 m	3/2(-)	+1.18(2)	A	ABLS	1985Co24	PL B163 66 (1985)
87 Fr 224	0	3.3 m	1(-)	+0.523(9)	A	ABLS	1985Co24	PL B163 66 (1985)
87 Fr 225	0	3.95 m	3/2-	+1.33(2)	A	ABLS	1985Co24 1987Co19	PL B163 66 (1985) NP A468 1 (1987)
87 Fr 226	0	49 s	1	-1.37(3)	A	ABLS	1985Co24	PL B163 66 (1985)
87 Fr 228	0	38 s	2-	+2.41(6)	A	ABLS	1985Co24	PL B163 66 (1985)

Radium	Efg calculated in 7s7p states of the Ra atom								
Reference isotope	88 Ra 209	0	4.8 s	5/2-	+0.39(4)	223Ra	CLS	1989Ne03	ZP D11 105 (1989)
	88 Ra 211	0	13s	5/2-	+0.46(4)	223Ra	CLS	1989Ne03	ZP D11 105 (1989)
	88 Ra 221	0	28 s	5/2-	+1.92(6)	223Ra	CLS	1989Ne03	ZP D11 105 (1989)
	88 Ra 223	0	11.43 d	3/2+	+1.22(3)		CLS	2008Py02 2018Ly01	Mol Phys 106 1965 (2008) PR C97 024309 (2018)
	88 Ra 227	0	42.2 m	3/2+	+1.53(6)	223Ra	CLS	1989Ne03	ZP D11 105 (1989)
	88 Ra 229	0	4.0 m	5/2(+)	+2.99(12)	223Ra	CLS	1989Ne03	ZP D11 105 (1989)

Actinium	<i>Calculated efg in states in the Ac atom</i>							2017Fe10	Nat Commun 8 14520 (2017)
	89 Ac 214	0	8.3 s	5+	+0.14(8)		IGLIS	2017Gr18	PR C96 054331 (2017)
	89 Ac 215	0	0.17 s	9/2-	+0.04(10)		IGLIS	2017Gr18	PR C96 054331 (2017)
	89 Ac 227	0	21.77 y	3/2-	+1.74(10)		O/R	1955Fr26 2017Fe10	PR 98 1514 and 111 1747/ Nat Commun 8 14520 (2017)
Thorium	<i>Efg calculated in states of Th 3+ ions</i>							2013Sa69	PR A88 060501 (2013)
	90 Th 229	0	7880 y	5/2+	+3.11(6)	A	O	2011Ca17	PRL 106 223001 (2011)
Protoactinium	<i>There is no adopted reference efg for Pa.</i>								
	<i>A. Estimated from B(E2) value.</i>								
	<i>B. Based on estimated efg in the Pr atom</i>								
	91 Pa 231	0	3.3 10 ⁴ y	3/2-	[-1.72(5)]	from B(E2)	not measured	1978Fr28	PL A69 225 (1975)
		84	45.1 ns	5/2+	+0.7(2)	231Pa	ME	1978Fr28	PL A69 225 (1975)
	91 Pa 233	0	27.0 d	3/2-	-3.0(4)	B	AB	1961Ma42	NP 23 90 (1961)
Uranium	<i>Muonic atom X-ray hyperfine structure</i>								
<i>Reference isotope</i>	92 U 233	0	1.6 10 ⁵ y	5/2+	+3.663(8)		Mu-X	1984Zu02	PRL 53 1888 (1984)
		40	0.11 ns	7/2+	+0.64(3)		Mu-X	1984Zu02	PRL 53 1888 (1984)
<i>Reference isotope</i>	92 U 235	0	7.0 10 ⁸ y	7/2-	+4.936(6)		Mu-X	1984Zu02	PRL 53 1888 (1984)
		46	~ 14 ps	9/2-	+1.87(3)		Mu-X	1984Zu02	PRL 53 1888 (1984)
Neptunium	<i>Muonic atom X-ray hyperfine structure</i>								
<i>Reference isotope</i>	93 Np 237	0	2.1 10 ⁶ y	5/2+	+3.886(6)		Mu-X	1987De10	PL B189 7 (1987)
		60	67 ns	5/2-	+3.85(4)	237Np	ME	1968Pi01 1968St03	BAPS 13 28 (1968) PR 165 1319 (1968)
Plutonium	<i>Muonic atom X-ray hyperfine structure</i>								
	<i>A. Calculated efg of the 8F3/2 state of Pu II</i>								
<i>Reference isotope</i>	94 Pu 239	8	36 ps	3/2+	-2.319(7)		Mu-X	1986Zu01	PL B167 383 (1986)
		57	101 ps	5/2+	-3.345(13)		Mu-X	1986Zu01	PL B167 383 (1986)
		76	111 ps	7/2+	-3.83(3)		Mu-X	1986Zu01	PL B167 383 (1986)
	94 Pu 241	0	14.3 y	5/2+	+6(2)	A	O	1964Ch12	JPPa 25 825 (1964)
Americium	<i>Muonic atom X-ray hyperfine structure</i>								
<i>Reference isotope</i>	95 Am 241	0	432.6 y	5/2-	+4.34(5)		Mu-X	1985Jo04	PL B161 75 (1985)
	95 Am 242	0	16.0 h	1-	-2.44((3)	241Am	AB	1966Ar04	PR 144 994 (1966)
		49	152 y	5-	+6.7(4)	241Am	ABLS	1988Be30	ZP A330 235 (1988)

	95 Am 243	0 84	7364 y 2.3 ns	5/2- 5/2+	+4.32(6) +4.2(2)	241Am	Mu-X ME	1985Jo04 1976Bo13	PL B161 75 (1985) JINC 38 1291 (1976)
Einsteinium	<i>Efg calculated in the Es atom</i>								
Reference isotope	99 Es 253	0	20.5 d	7/2+	+6.7(8)		AB	1975Go05	PR A11 499 (1975)
Reference isotope	99 Es 254	84	39.3 h	2+	+3.7(5)		AB	1975Go05	PR A11 499 (1975)
Nobelium	<i>Efg calculated in the 1P1 level of the No atom</i>								
	102 No 253	0	1.62 m	(9/2-)	+5.9(17)		CLS	2018Ra11	PRL 120 232503 (2018)

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